# Statistics

## Chapter 11 — Instructor Solutions Manual

October 16, 2026 • Total Exercises: 65

**PROBLEM STATEMENT**

Chapter 11, Page 377

Step 1

Given:

$$
\begin{aligned}
H_0 &: \mu_1 = \mu_2 \\
H_1 &: \mu_1 \neq \mu_2 \\
n_1 &= \text{Sample size} = 40 \\
n_2 &= \text{Sample size} = 50 \\
\overline{x}_1 &= \text{Sample mean} = 102 \\
\overline{x}_2 &= \text{Sample mean} = 99 \\
\sigma_1 &= \text{Population standard deviation} = 5 \\
\sigma_2 &= \text{Population standard deviation} = 6 \\
\alpha &= \text{Significance level} = 0.04
\end{aligned}
$$

(a) If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed (one-tailed).

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed (one-tailed).

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed test}$$

Step 2

(b) The rejection region of a two-tailed test with $\alpha = 0.04$ contains all z-values below the z-value $-z_0$ that has a probability of $0.04/2 = 0.02$ to its left and all z-values above the z-value $z_0$ that has a probability of $0.04/2 = 0.02$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.04}{2} = 0.02$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.02 = 0.48$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.48 is closest to 0.4798, which is given in the row starting with "2.0" and in the column starting with ".05", thus the corresponding z-score is $2.0 + .05 = 2.05$.

$$z_0 = \pm 2.05$$

Thus the rejection region then contains all z-scores smaller than $-2.05$ and all z-scores larger than 2.05, thus we reject $H_0$ when the z-score is smaller than $-2.05$ or the z-score is larger than 2.05.

Step 3

(c) Determine the value of the test statistic:

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{102 - 99 - 0}{\sqrt{\frac{5^2}{40} + \frac{6^2}{50}}} \approx 2.59$$

Step 4

(d) If the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$.

$$2.59 > 2.05 \Rightarrow \text{ Reject } H_0$$

Step 5

(e)

## P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic $z$, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -2.59$ or larger than $z = 2.59$.

$$P(z < -2.59 \text{ or } z > 2.59)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 2.59)$ is given in the row starting with "$2.5$" and in the column starting with ".09" of the table.

$$\begin{aligned} P &= P(z < -2.59 \text{ or } z > 2.59) \\ &= 2P(z > 2.59) \\ &= 2(1 - P(z < 2.59)) \\ &= 2(1 - P(z < 0) - P(0 < z < 2.59)) \\ &= 2(1 - 0.5000 - 0.4952) \\ &= 2(0.0048) \\ &= 0.0096 \\ &= 0.96\% \end{aligned}$$

Result

(a) Two-tailed test

(b) Reject $H_0$ when $z < -2.05$ or $z > 2.05$

(c) $z = 2.59$

(d) Reject $H_0$

(e) $P = 0.0096$

**PROBLEM STATEMENT**

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**Step 1**

Given:

$$\begin{aligned} H_0 &: \mu_1 \leq \mu_2 \\ H_1 &: \mu_1 > \mu_2 \\ n_1 &= \text{Sample size} = 65 \\ n_2 &= \text{Sample size} = 50 \\ \overline{x}_1 &= \text{Sample mean} = 2.67 \\ \overline{x}_2 &= \text{Sample mean} = 2.59 \\ \sigma_1 &= \text{Population standard deviation} = 0.75 \\ \sigma_2 &= \text{Population standard deviation} = 0.66 \\ \alpha &= \text{Significance level} = 0.08 \end{aligned}$$

(a) If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed (one-tailed).

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed (one-tailed).

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

$$\text{One-tailed test}$$

**Step 2**

(b) The rejection region of a right-tailed test with $\alpha = 0.08$ contains all z-scores above the z-score $z_0$ that has a probability of 0.08 to its right.

$$P(z > z_0) = 0.08$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.08 = 0.42$ in the normal probability table of the appendix. We note that the probability 0.42 is closest to 0.4207, which is given in the row starting with "1.4" and in the column starting with ".01", thus the corresponding z-score is $1.4 + .01 = 1.41$.

$$z_0 = 1.41$$

Thus the rejection region then contains all z-scores greater than $1.41$ and thus we reject $H_0$ when the z-score is greater than 1.41.

Step 3

(c) Determine the value of the test statistic:

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{2.67 - 2.59 - 0}{\sqrt{\frac{0.75^2}{65} + \frac{0.66^2}{50}}} \approx 0.61$$

Step 4

(d) If the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$.

$$0.61 < 1.41 \Rightarrow \text{ Fail to reject } H_0$$

Step 5

(e)

## P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic $z$, assuming that the null hypothesis is true.

Since the test is right-tailed, the P-value is the probability that the z-score is larger than $z = 0.61$.

$$P(z > 0.61)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 0.61)$ is given in the row starting with "$0.6$" and in the column starting with ".01" of the table.

$$\begin{aligned} P &= P(z > 0.61) \\ &= 1 - P(z < 0.61) \\ &= 1 - P(z < 0) - P(0 < z < 0.61) \\ &= 1 - 0.5000 - 0.2291 \\ &= 0.2709 \\ &= 27.09\% \end{aligned}$$

🎯 **Result**

(a) One-tailed test

(b) Reject $H_0$ when $z > 1.41$

(c) $z = 0.61$

(d) Fail to reject $H_0$

(e) $P = 0.2709$

**PROBLEM STATEMENT**

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Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 40 \\
n_2 &= \text{Sample size} = 55 \\
\overline{x}_1 &= \text{Sample mean} = 7.6 \\
\overline{x}_2 &= \text{Sample mean} = 8.1 \\
\sigma_1 &= \text{Population standard deviation} = 2.3 \\
\sigma_2 &= \text{Population standard deviation} = 2.9 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

Given claim: Mean is less for Gibbs brand

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$
\begin{aligned}
H_0 &: \mu_1 \geq \mu_2 \\
H_a &: \mu_1 < \mu_2
\end{aligned}
$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Left-tailed test

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all z-scores below the z-score $z_0$ that has a probability of 0.05 to its left.

$$P(z < z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = -1.65$$

Thus the rejection region then contains all z-scores smaller than $-1.65$ and thus we reject $H_0$ when the z-score is smaller than $-1.65$.

Determine the value of the test statistic:

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{7.6 - 8.1 - 0}{\sqrt{\frac{2.3^2}{40} + \frac{2.9^2}{55}}} \approx -0.94$$

If the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$.

$$-0.94 > -1.65 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that babies using the Gibbs brand gained less weight.

Step 3

### P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic $z$, assuming that the null hypothesis is true.

Since the test is left-tailed, the P-value is the probability that the z-score is smaller than $z = -0.94$.

$$P(z < -0.94)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 0.94)$ is given in the row starting with "$0.9$" and in the column starting with ".04" of the table.

$$\begin{aligned} P &= P(z < -0.94) \\ &= P(z > 0.94) \\ &= 1 - P(z < 0.94) \\ &= 1 - P(z < 0) - P(0 < z < 0.94) \\ &= 1 - 0.5000 - 0.3264 \\ &= 0.1736 \\ &= 17.36\% \end{aligned}$$

Result

No, $P = 0.1736$

**PROBLEM STATEMENT**

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Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 35 \\
n_2 &= \text{Sample size} = 40 \\
\overline{x}_1 &= \text{Sample mean} = 370 \\
\overline{x}_2 &= \text{Sample mean} = 380 \\
\sigma_1 &= \text{Population standard deviation} = 30 \\
\sigma_2 &= \text{Population standard deviation} = 26 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

Given claim: Difference in means

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$
\begin{aligned}
H_0 &: \mu_1 = \mu_2 \\
H_a &: \mu_1 \neq \mu_2
\end{aligned}
$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Two-tailed test

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all z-values below the z-value $-z_0$ that has a probability of $0.05/2 = 0.025$ to its left and all z-values above the z-value $z_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.025 = 0.475$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.475 is closest to 0.4750, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = \pm 1.96$$

Thus the rejection region then contains all z-scores smaller than $-1.96$ and all z-scores larger than 1.96, thus we reject $H_0$ when the z-score is smaller than $-1.96$ or the z-score is larger than 1.96.

Determine the value of the test statistic:

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{370 - 380 - 0}{\sqrt{\frac{30^2}{35} + \frac{26^2}{40}}} \approx -1.53$$

If the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$.

$$-1.96 < -1.53 < 1.96 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a different in the mean number of miles traveled per month between Cincinnati and Pittsburgh employees.

Result

No

**PROBLEM STATEMENT**

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Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 45 \\
n_2 &= \text{Sample size} = 39 \\
\overline{x}_1 &= \text{Sample mean} = 61.4 \\
\overline{x}_2 &= \text{Sample mean} = 60.6 \\
\sigma_1 &= \text{Population standard deviation} = 1.2 \\
\sigma_2 &= \text{Population standard deviation} = 1.1 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

Given claim: Mean is less for Caesarean section (shorter)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$
\begin{aligned}
H_0 &: \mu_1 \leq \mu_2 \\
H_a &: \mu_1 > \mu_2
\end{aligned}
$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score $z_0$ that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than $1.65$ and thus we reject $H_0$ when the z-score is greater than 1.65.

Determine the value of the test statistic:

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{61.4 - 60.6 - 0}{\sqrt{\frac{1.2^2}{45} + \frac{1.1^2}{39}}} \approx 3.19$$

If the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$.

$$3.19 > 1.65 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that those who had a Caesarean section are shorter.

Step 3

### P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic $z$, assuming that the null hypothesis is true.

Since the test is right-tailed, the P-value is the probability that the z-score is larger than $z = 3.19$.

$$P(z > 3.19)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 3.19)$ is given in the row starting with "$3.1$" and in the column starting with ".09" of the table.

$$\begin{aligned} P &= P(z > 3.19) \\ &= 1 - P(z < 3.19) \\ &= 1 - P(z < 0) - P(0 < z < 3.19) \\ &= 1 - 0.5000 - 0.4993 \\ &= 0.0007 \\ &= 0.07\% \end{aligned}$$

Result

Yes, $P = 0.0007$

**PROBLEM STATEMENT**

Chapter 11, Page 378

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 40 \\ n_2 &= \text{Sample size} = 45 \\ \overline{x}_1 &= \text{Sample mean} = 20.75 \\ \overline{x}_2 &= \text{Sample mean} = 19.80 \\ \sigma_1 &= \text{Population standard deviation} = 2.25 \\ \sigma_2 &= \text{Population standard deviation} = 1.90 \\ \alpha &= \text{Significance level} = 0.02 \end{aligned}$$

Given claim: Mean is larger for union (earn more)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$\begin{aligned} H_0 &: \mu_1 \leq \mu_2 \\ H_a &: \mu_1 > \mu_2 \end{aligned}$$

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.02$ contains all z-scores above the z-score $z_0$ that has a probability of 0.02 to its right.

$$P(z > z_0) = 0.02$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.02 = 0.48$ in the normal probability table of the appendix. We note that the probability 0.48 is closest to 0.4798, which is given in the row starting with "2.05" and in the column starting with ".05", thus the corresponding z-score is $2.0 + .05 = 2.05$.

$$z_0 = 2.05$$

Thus the rejection region then contains all z-scores greater than $2.05$ and thus we reject $H_0$ when the z-score is greater than 2.05.

Step 2

Determine the value of the test statistic:

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{20.75 - 19.80 - 0}{\sqrt{\frac{2.25^2}{40} + \frac{1.90^2}{45}}} \approx 2.09$$

If the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$.

$$2.09 > 2.05 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that union nurses earn more.

Step 3

### P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic $z$, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is larger than $z = 2.09$.

$$P(z > 2.09)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 2.09)$ is given in the row starting with "$2.0$" and in the column starting with ".09" of the table.

$$\begin{aligned} P &= P(z > 2.09) \\ &= 1 - P(z < 2.09) \\ &= 1 - P(z < 0) - P(0 < z < 2.09) \\ &= 1 - 0.5000 - 0.4817 \\ &= 0.0183 \\ &= 1.83\% \end{aligned}$$

Result

Yes, $P = 0.0183$

**PROBLEM STATEMENT**

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Step 1

Given:

$$
\begin{aligned}
H_0 &: \pi_1 \leq \pi_2 \\
H_1 &: \pi_1 > \pi_2 \\
n_1 &= \text{Sample size} = 100 \\
n_2 &= \text{Sample size} = 150 \\
x_1 &= \text{Number of successes} = 70 \\
x_2 &= \text{Number of successes} = 90 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

(a) If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed test}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score $z_0$ that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than $1.65$ and thus we reject $H_0$ when the z-score is greater than 1.65.

Step 2

(b) The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{70}{100} \approx 0.70$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{90}{150} \approx 0.60$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{70 + 90}{100 + 150} = \frac{160}{250} \approx 0.64$$

Step 3

(c) Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.70 - 0.60}{\sqrt{0.64(1-0.64)}\sqrt{\frac{1}{100} + \frac{1}{150}}} \approx 1.61$$

Step 4

(d) When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$z = 1.61 < 1.65 \Rightarrow \text{ Fail to reject } H_0$$

Result

(a) Reject $H_0$ when $z > 1.65$

(b) $\hat{p}_c = 0.64$

(c) $z = 1.61$

(d) Fail to reject $H_0$

**PROBLEM STATEMENT**

Chapter 11, Page 381

Step 1

Given:

$$
\begin{aligned}
H_0 &: \pi_1 = \pi_2 \\
H_a &: \pi_1 \neq \pi_2 \\
n_1 &= \text{Sample size} = 200 \\
n_2 &= \text{Sample size} = 150 \\
x_1 &= \text{Number of successes} = 170 \\
x_2 &= \text{Number of successes} = 110 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

(a) If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed test}$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all z-values below the z-value $-z_0$ that has a probability of $0.05/2 = 0.025$ to its left and all z-values above the z-value $z_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.025 = 0.475$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.475 is closest to 0.4750, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = \pm 1.96$$

Thus the rejection region then contains all z-scores smaller than $-1.96$ and all z-scores larger than 1.96, thus we reject $H_0$ when the z-score is smaller than $-1.96$ or the z-score is larger than 1.96.

Step 2

(b) The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{170}{200} \approx 0.85$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{110}{150} \approx 0.7333$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{170 + 110}{200 + 150} = \frac{280}{350} \approx 0.8$$

Step 3

(c) Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.85 - 0.7333}{\sqrt{0.8(1 - 0.8)}\sqrt{\frac{1}{200} + \frac{1}{150}}} \approx 2.70$$

Step 4

(d) When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$z = 2.70 > 1.96 \Rightarrow \text{ Reject } H_0$$

Result

(a) Reject $H_0$ when $z < -1.96$ or $z > 1.96$

(b) $\hat{p}_c = 0.8$

(c) $z = 2.70$

(d) Reject $H_0$

**PROBLEM STATEMENT**

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Step 1

Given:

$$n_1 = \text{Sample size} = 400$$
$$n_2 = \text{Sample size} = 400$$
$$x_1 = \text{Number of successes} = 24$$
$$x_2 = \text{Number of successes} = 40$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Difference in the proportions

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi_1 = \pi_2$$
$$H_a : \pi_1 \neq \pi_2$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Two-tailed test

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all z-values below the z-value $-z_0$ that has a probability of $0.05/2 = 0.025$ to its left and all z-values above the z-value $z_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.025 = 0.475$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.475 is closest to 0.4750, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = \pm 1.96$$

Thus the rejection region then contains all z-scores smaller than $-1.96$ and all z-scores larger than 1.96, thus we reject $H_0$ when the z-score is smaller than $-1.96$ or the z-score is larger than 1.96.

Step 3

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{24}{400} \approx 0.06$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{40}{400} \approx 0.1$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{24 + 40}{400 + 400} = \frac{64}{800} \approx 0.08$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.06 - 0.1}{\sqrt{0.08(1 - 0.08)}\sqrt{\frac{1}{400} + \frac{1}{400}}} \approx -2.09$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$z = -2.09 < -1.96 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is a different in the proportion of vines infested using Pernod 5 as opposed to Action.

🎯 Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 382

Step 1

Given:

$$n_1 = \text{Sample size} = 3000$$
$$n_2 = \text{Sample size} = 3000$$
$$x_1 = \text{Number of successes} = 2010$$
$$x_2 = \text{Number of successes} = 1530$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Proportion is less for the later survey

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi_1 \leq \pi_2$$
$$H_a : \pi_1 > \pi_2$$

**Step 2**

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score $z_0$ that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than $1.65$ and thus we reject $H_0$ when the z-score is greater than 1.65.

**Step 3**

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{2010}{3000} \approx 0.67$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{1530}{3000} \approx 0.51$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{2010 + 1530}{3000 + 3000} = \frac{3540}{6000} \approx 0.59$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.67 - 0.51}{\sqrt{0.59(1 - 0.59)}\sqrt{\frac{1}{3000} + \frac{1}{3000}}} \approx 12.60$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$z = 12.60 > 1.65 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that women think men are less kind, gentle, and thoughtful in the later survey compared with the earlier one.

🎯 Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 382

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 1000 \\
n_2 &= \text{Sample size} = 800 \\
x_1 &= \text{Number of successes} = 200 \\
x_2 &= \text{Number of successes} = 168 \\
\alpha &= \text{Significance level} = 0.02
\end{aligned}
$$

Given claim: Proportion is larger for Democrats

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$
\begin{aligned}
H_0 &: \pi_1 \geq \pi_2 \\
H_a &: \pi_1 < \pi_2
\end{aligned}
$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Left-tailed test

The rejection region of a left-tailed test with $\alpha = 0.02$ contains all z-scores below the z-score $z_0$ that has a probability of 0.02 to its left.

$$P(z < z_0) = 0.02$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.02 = 0.48$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.48 is closest to 0.4798, which is given in the row starting with "2.0" and in the column starting with ".05", thus the corresponding z-score is $2.0 + .05 = 2.05$.

$$z_0 = -2.05$$

Thus the rejection region then contains all z-scores smaller than $-2.05$ and thus we reject $H_0$ when the z-score is smaller than $-2.05$.

Step 3

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{200}{1000} \approx 0.2$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{168}{800} \approx 0.21$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{200 + 168}{1000 + 800} = \frac{368}{1800} \approx 0.2044$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.2 - 0.21}{\sqrt{0.2044(1 - 0.2044)}\sqrt{\frac{1}{1000} + \frac{1}{800}}} \approx -0.52$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$z = -0.52 > -2.05 \Rightarrow \text{ Fail to reject } H_0$$

There is sufficient evidence to support the claim that there is a larger proportion of Democrats in favor of lowering the standards.

Step 4

### P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic $z$, assuming that the null hypothesis is true.

Since the test is left-tailed, the P-value is the probability that the z-score is smaller than $z = -0.52$.

$$P(z < -0.52)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 0.52)$ is given in the row starting with "$0.5$" and in the column starting with ".02" of the table.

$$\begin{aligned} P &= P(z < -0.52) \\ &= P(z > 0.52) \\ &= 1 - P(z < 0.52) \\ &= 1 - P(z < 0) - P(0 < z < 0.52) \\ &= 1 - 0.5000 - 0.1985 \\ &= 0.3015 \\ &= 30.15\% \end{aligned}$$

🎯 **Result**

No, $P = 0.3015$

**PROBLEM STATEMENT**

Chapter 11, Page 382

Step 1

Given:

$$n_1 = \text{Sample size} = 400$$
$$n_2 = \text{Sample size} = 600$$
$$x_1 = \text{Number of successes} = 120$$
$$x_2 = \text{Number of successes} = 150$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Difference in the proportions

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi_1 = \pi_2$$
$$H_a : \pi_1 \neq \pi_2$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Two-tailed test

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all z-values below the z-value $-z_0$ that has a probability of $0.05/2 = 0.025$ to its left and all z-values above the z-value $z_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.025 = 0.475$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.475 is closest to 0.4750, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = \pm 1.96$$

Thus the rejection region then contains all z-scores smaller than $-1.96$ and all z-scores larger than 1.96, thus we reject $H_0$ when the z-score is smaller than $-1.96$ or the z-score is larger than 1.96.

Step 3

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{120}{400} \approx 0.3$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{150}{600} \approx 0.25$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{120 + 150}{400 + 600} = \frac{270}{1000} \approx 0.27$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.3 - 0.25}{\sqrt{0.27(1 - 0.27)}\sqrt{\frac{1}{400} + \frac{1}{600}}} \approx 1.74$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-1.96 < 1.74 < 1.96 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that there is a different in the proportions of single and married persons having an accident during a three-year period.

Step 4

### P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic $z$, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -1.74$ or larger than $z = 1.74$.

$$P(z < -1.74 \text{ or } z > 1.74)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.74)$ is given in the row starting with "$1.7$" and in the column starting with ".04" of the table.

$$\begin{aligned} P &= P(z < -1.74 \text{ or } z > 1.74) \\ &= 2P(z > 1.74) \\ &= 2(1 - P(z < 1.74)) \\ &= 2(1 - P(z < 0) - P(0 < z < 1.74)) \\ &= 2(1 - 0.5000 - 0.4591) \\ &= 2(0.0409) \\ &= 0.0818 \\ &= 8.18\% \end{aligned}$$

Result

No, $P = 0.0818$

**PROBLEM STATEMENT**

Chapter 11, Page 387

**Solution A**

**Step 1**

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 10 \\
n_2 &= \text{Sample size} = 8 \\
\overline{x}_1 &= \text{Sample mean} = 23 \\
\overline{x}_2 &= \text{Sample mean} = 26 \\
s_1 &= \text{Sample standard deviation} = 4 \\
s_2 &= \text{Sample standard deviation} = 5 \\
\alpha &= \text{Significance level} = 0.05 \\
H_0 &: \mu_1 = \mu_2 \\
H_1 &: \mu_1 \neq \mu_2
\end{aligned}
$$

(a) If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 10 + 8 - 2 = 16$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.120$$

The rejection region then contains all values smaller than $-2.120$ and all values greater than $2.120$, thus we reject $H_0$ when $t < -2.120$ or $t > 2.120$.

Step 2

(b) Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(10 - 1)4^2 + (8 - 1)5^2}{10 + 8 - 2}} \approx 4.4651$$

Step 3

(c) Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{23 - 26}{4.4651\sqrt{\frac{1}{10} + \frac{1}{8}}} \approx -1.416$$

Step 4

(d) When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-2.120 < -1.416 < 2.120 \Rightarrow \text{ Fail to reject } H_0$$

Step 5

(e)

## P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value $1.416$ in the row $df = n_1 + n_2 - 2 = 10 + 8 - 2 = 16$ (two-tailed):

$$0.10 < P < 0.20$$

*Note: When using technology, then you obtain the P-value 0.17594.*

**Result**

(a) Reject $H_0$ when $t < -2.120$ or $t > 2.120$

(b) $s_p = 4.4651$

(c) $t = -1.416$

(d) Fail to reject $H_0$

(e) $0.10 < P < 0.20$

**Solution B**

**Step 1**

**a.)** Recall that two-tailed test should be used when there's an equality sign in the hypothesis and one-tailed test when the hypothesis has an inequality sign. Since the given hypothesis has an equality sign, then it should use

$$\text{two-tailed test}$$

where the rejection region will be on the both tail of the of the curve.

**Step 2**

Since it's a two-tailed test, the significance level $\alpha$ of each tail will be

$$\alpha = \frac{0.05}{2} = 0.025.$$

which will cover the rejection region in the left and the right tail of the curve. To find the critical values, it needed to find the degree of freedom by using the formula

$$df = n_1 + n_2 - 2.$$

Substituting the given sample observations into the formula of the degree of freedom, thus

$$\begin{aligned} df &= n_1 + n_2 - 2 = 10 + 8 - 2 \\ &= 16 \end{aligned}$$

**Step 3**

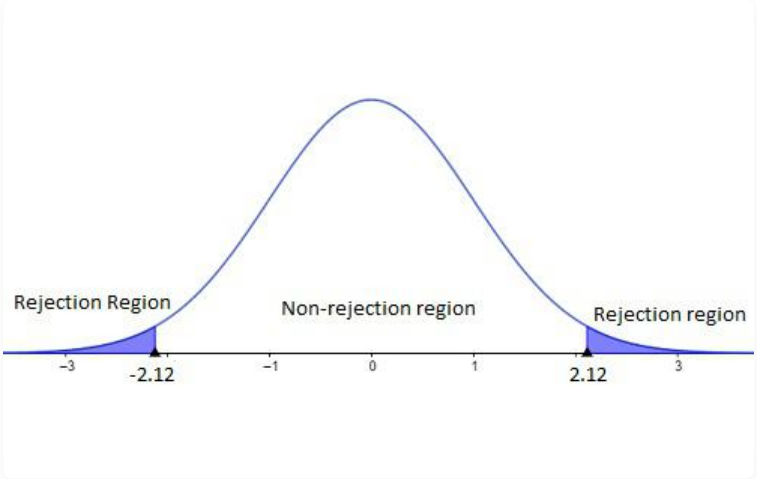
Using a $T$-table to find the critical value where the row is the degree of freedom $16$ and the column is the significance level $\alpha$ as two-tailed $0.05$, it will be

| $df/\alpha$ | $\cdots$ | $0.05$ | $\cdots$ |
|---|---|---|---|
| $\vdots$ | $\ddots$ | $\vdots$ | |
| $16$ | $\cdots$ | $2.120$ | |
| $\vdots$ | | | |

By observing the table, the critical value where the row and the column intersects is at

$$t = 2.120$$

To visualize the critical value, the graph is represented below.




**Step 4**

Since the rejection region is on the both tails of the curve, then fail to reject the null hypothesis when $-2.120 < t < 2.120$. The decision rule will be

$$\text{When } t < -2.120 \text{ or } t > 2.120, \text{ reject } H_0$$

Step 5

**b.)** Since the given is only a sample standard deviation, it needed to use the formula

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

to determine the pooled estimate of the pooled variance where:

- $n_1$ and $n_2$ are sample size,
- $s_1$ and $s_2$ are sample standard deviation, and
- $s_p^2$ is the pooled estimate of the pooled variance.

Step 6

By substituting when sample size $n_1$ is $10$, sample size $n_2$ is $8$, estimate standard deviation $s_1$ is $4$ , and estimate standard deviation $s_2$ is $5$ into the formula, thus

$$\begin{aligned} s_p^2 &= \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \\ &= \frac{(10 - 1)(4)^2 + (8 - 1)(5)^2}{10 + 8 - 2} \\ &= 19.9375 \end{aligned}$$

In conclusion, the pooled estimate of the pooled variance is

$$s_p^2 = 19.9375$$

Step 7

**c.)** The test statistic that will be using is a two-sample test of means where the population standard deviation is unknown

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where:

- $\bar{x}_1$ and $\bar{x}_2$ are sample means,
- $s_p^2$ is the pooled estimate of the pooled variance,
- $n_1$ and $n_2$ are sample size, and
- $t$ is the test statistics.

The two-sample test of means where the population standard deviation is unknown should be used since there's two independent parameters that wants to be compared and two given estimate standard deviations.

Step 8

Substituting the sample observations $n_1$ is $10$ and $n_2$ is $8$, sample means $\bar{x}_1$ is $23$ and $\bar{x}_2$ is $26$, sample standard deviation $s_1$ is $4$ and $s_2$ is $5$, and the pooled estimate of the pooled variance $s_p^2$ is $19.9375$ into the formula to find the test statistics of two means where the population standard $\sigma$ is unknown, thus

$$\begin{aligned} t &= \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \\ &= \frac{23 - 26}{\sqrt{19.9375\left(\frac{1}{10} + \frac{1}{8}\right)}} \\ &= -1.416428464 \approx -1.4164. \end{aligned}$$

In conclusion, the test statistics of the given hypothesis is

$$t = -1.4164$$

Step 9

$\mathbf{d.)}$ Since the test statistic $-1.4164$ is greater than the critical value of $-2.120$ and less than $2.120$, then the null hypothesis is fail to be rejected. The decision will be

- At $0.05$ significance level **fail** to reject $H_0$.

Step 10

$\mathbf{e.)}$ The $p$-value or probability value represents the probability of how likely it is to obtain the value of the test statistics given that the null hypothesis is true. To estimate the $p$-value, Assume that $H_0$ is true. Use the $T$-table where the row is when degree of freedom is $16$ and the closest value obtained from test statistics which is the absolute value of $-1.416$. The table will be

| $df/\alpha$ | $\cdots$ | $0.20$ | $0.10$ | $\cdots$ |
|---|---|---|---|---|
| $\vdots$ | $\ddots$ | $\vdots$ | $\vdots$ | |
| $16$ | $\cdots$ | $1.337$ | $1.746$ | |
| $\vdots$ | | | | |

By observing the table, when the significance level is from $0.10$ and $0.20$, the test statistics is within its test values

$$1.337 < |-1.416| < 1.746$$

Step 11

The column where the test statistics is closest is when the significance level is from $0.10$ to $0.20$. Hence, this will be the upper bound and the lower bound limit of the estimates. In conclusion, the estimate of the $p$-value will be

$$0.10 < P < 0.20$$

🎯 **Result**

*a.*) When $t < -2.120$ or $t > 2.120$, reject $H_0$

*b.*) $s_p^2 = 19.9375$

*c.*) $t = -1.4164$

*d.*) At 0.05 significance level, fail to reject $H_0$.

*e.*) $0.10 < P < 0.20$

**PROBLEM STATEMENT**

Chapter 11, Page 387

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 15 \\
n_2 &= \text{Sample size} = 17 \\
\overline{x}_1 &= \text{Sample mean} = 350 \\
\overline{x}_2 &= \text{Sample mean} = 342 \\
s_1 &= \text{Sample standard deviation} = 12 \\
s_2 &= \text{Sample standard deviation} = 15 \\
\alpha &= \text{Significance level} = 0.10 \\
H_0 &: \mu_1 = \mu_2 \\
H_1 &: \mu_1 \neq \mu_2
\end{aligned}
$$

(a) If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.10$ contains all t-values below the t-value $-t_0$ that has a probability of $0.10/2 = 0.05$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.10/2 = 0.05$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.10}{2} = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 15 + 17 - 2 = 30$ and in the column with $\alpha = 0.10$ (two-tailed).

$$t = 1.697$$

The rejection region then contains all values smaller than $-1.697$ and all values greater than $1.697$, thus we reject $H_0$ when $t < -1.697$ or $t > 1.697$.

Step 2

(b) Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(15 - 1)12^2 + (17 - 1)15^2}{15 + 17 - 2}} \approx 13.6821$$

Step 3

(c) Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{350 - 342}{13.6821\sqrt{\frac{1}{15} + \frac{1}{17}}} \approx 1.651$$

Step 4

(d) When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-1.697 < 1.651 < 1.697 \Rightarrow \text{ Fail to reject } H_0$$

Step 5

(e)

## P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value $1.651$ in the row $df = n_1 + n_2 - 2 = 15 + 17 - 2 = 30$ (two-tailed):

$$0.10 < P < 0.20$$

*Note: When using technology, then you obtain the P-value 0.10918.*

🎯 **Result**

(a) Reject $H_0$ when $t < -1.697$ or $t > 1.697$

(b) $s_p = 13.6821$

(c) $t = 1.651$

(d) Fail to reject $H_0$

(e) $0.10 < P < 0.20$

**PROBLEM STATEMENT**

Chapter 11, Page 387

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 12 \\ n_2 &= \text{Sample size} = 13 \\ \alpha &= \text{Significance level} = 0.01 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{453.7 + 16500 + ... + 11500}{12} = 7240.2583 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{9000 + 410.8 + ... + 1100}{13} = 12996.1846 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n-1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(453.7-7240.2583)^2 + ... + (11500-7240.2583)^2}{12-1}} \\ &= 8242.2084 \\ s_2 &= \sqrt{\frac{\sum (x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(9000-12996.1846)^2 + .... + (1100-12996.1846)^2}{13-1}} \\ &= 16296.1802 \end{aligned}$$

Step 2

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.01$ contains all t-values below the t-value $-t_0$ that has a probability of $0.01/2 = 0.005$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.01/2 = 0.005$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.01}{2} = 0.005$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 12 + 13 - 2 = 23$ and in the column with $\alpha = 0.01$ (two-tailed).

$$t = 3.104$$

The rejection region then contains all values smaller than $-3.104$ and all values greater than $3.104$,thus we reject $H_0$ when $t < -3.104$ or $t > 3.104$.

Step 3

Determine the pooled standard deviation:

$$\begin{aligned} s_p &= \sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}} \\ &= \sqrt{\frac{(12-1)8242.2084^2+(13-1)16296.1802^2}{12+13-2}} \\ &\approx 13078.45866 \end{aligned}$$

Determine the test statistic:

$$t = \frac{\overline{x}_1-\overline{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{7240.2583-12996.1846}{13078.45866\sqrt{\frac{1}{12}+\frac{1}{13}}} \approx -1.099$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-3.104 < -1.099 < 3.104 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean salaries between position players and pitchers are the same.

Result

There is not sufficient evidence to support the claim that the mean salaries between position players and pitchers are the same.

**PROBLEM STATEMENT**

Chapter 11, Page 387

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 15 \\
n_2 &= \text{Sample size} = 12 \\
\overline{x}_1 &= \text{Sample mean} = 61 \\
\overline{x}_2 &= \text{Sample mean} = 48.4 \\
s_1 &= \text{Sample standard deviation} = 15.5 \\
s_2 &= \text{Sample standard deviation} = 18.1 \\
\alpha &= \text{Significance level} = 0.01
\end{aligned}
$$

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.01$ contains all t-values below the t-value $-t_0$ that has a probability of $0.01/2 = 0.005$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.01/2 = 0.005$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.01}{2} = 0.005$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 15 + 12 - 2 = 25$ and in the column with $\alpha = 0.01$ (two-tailed).

$$t = 3.078$$

The rejection region then contains all values smaller than $-3.078$ and all values greater than $3.078$, thus we reject $H_0$ when $t < -3.078$ or $t > 3.078$.

Step 2

Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(15 - 1)15.5^2 + (12 - 1)18.1^2}{15 + 12 - 2}} \approx 16.6940$$

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{61 - 48.4}{16.6940\sqrt{\frac{1}{15} + \frac{1}{12}}} \approx 1.949$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-3.078 < 1.949 < 3.078 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the single-earner couples on average spend more time watching television together.

Result

No

**PROBLEM STATEMENT**

Chapter 11, Page 388

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 6 \\ n_2 &= \text{Sample size} = 7 \\ \alpha &= \text{Significance level} = 0.10 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{131 + 135 + ... + 142}{6} = 142.5 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{130 + 102 + ... + 139}{7} = 130.2857 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(131 - 142.5)^2 + .... + (142 - 142.5)^2}{6 - 1}} \\ &= 12.2434 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(130 - 130.2857)^2 + ... + (139 - 130.2857)^2}{7 - 1}} \\ &= 15.7873 \end{aligned}$$

Step 2

Given claim: Mean is greater for sales staff

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 \leq \mu_2$$

$$H_1 : \mu_1 > \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.10$ contains all t-value above the t-value $t_0$ that has a probability of 0.10 to its right.

$$P(t > t_0) = 0.10$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 6 + 7 - 2 = 11$ and in the column with $\alpha = 0.10$.

$$t = 1.796$$

The rejection region then contains all values greater than $1.796$ and thus we reject $H_0$ when the t-value is larger than 1.796.

Step 3

Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}} = \sqrt{\frac{(6-1)12.2434^2 + (7-1)15.7873^2}{6+7-2}} \approx 14.2858$$

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{142.5 - 130.2857}{14.2858\sqrt{\frac{1}{6}+\frac{1}{7}}} \approx 1.537$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$1.537 < 1.796 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean daily expenses are greater for the sales staff than the audit staff.

Step 4

### P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value $1.897$ in the row $df = n_1 + n_2 - 2 = 6 + 7 - 2 = 11$ (one-tailed):

$$0.05 < P < 0.10$$

Note: When using technology, then you obtain the P-value $0.07627$.

Result

No, $0.05 < P < 0.10$

**PROBLEM STATEMENT**

Chapter 11, Page 388

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 12 \\ n_2 &= \text{Sample size} = 8 \\ \alpha &= \text{Significance level} = 0.01 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{845 + 826 + \ldots + 832}{12} = 826.75 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{841 + 890 + \ldots + 829}{8} = 835.75 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n-1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(845 - 826.75)^2 + \ldots + (832 - 826.75)^2}{12-1}} \\ &= 22.8399 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(841 - 835.75)^2 + \ldots + (829 - 835.75)^2}{8-1}} \\ &= 34.4041 \end{aligned}$$

Step 2

Given claim: The mean is higher for nurses

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 \geq \mu_2$$

$$H_1 : \mu_1 < \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.10$ contains all t-value below the t-value $-t_0$ that has a probability of 0.10 to its left.

$$P(t < -t_0) = 0.10$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 12 + 8 - 2 = 18$ and in the column with $\alpha = 0.01$ (which is the negation of the critical value in the table).

$$t = -2.552$$

The rejection region then contains all values smaller than $-2.552$ and thus we reject $H_0$ when the t-value is smaller than $-2.552$.

Step 3

Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}} = \sqrt{\frac{(12-1)22.8399^2 + (8-1)34.4041^2}{12+8-2}} \approx 27.9123$$

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{826.75 - 835.75}{27.9123\sqrt{\frac{1}{12}+\frac{1}{8}}} \approx -0.706$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -0.706 > -2.552 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the mean weekly salary of nurses is higher.

Step 4

### P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value $0.706$ in the row $df = n_1 + n_2 - 2 = 12 + 8 - 2 = 18$ (one-tailed):

$$P > 0.10$$

Note: When using technology, then you obtain the P-value 0.24461.

Result

Not reasonable, $P > 0.10$

**PROBLEM STATEMENT**

Chapter 11, Page 391

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 15 \\
n_2 &= \text{Sample size} = 12 \\
\overline{x}_1 &= \text{Sample mean} = 50 \\
\overline{x}_2 &= \text{Sample mean} = 46 \\
s_1 &= \text{Sample standard deviation} = 5 \\
s_2 &= \text{Sample standard deviation} = 15 \\
\alpha &= \text{Significance level} = 0.05 \\
H_0 &: \mu_1 = \mu_2 \\
H_1 &: \mu_1 \neq \mu_2
\end{aligned}
$$

(a) Determine the degrees of freedom (rounded down to the nearest integer):

$$
\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{5^2}{15} + \frac{15^2}{12}\right)^2}{\frac{(5^2/15)^2}{15 - 1} + \frac{(15^2/12)^2}{12 - 1}} \approx 12
$$

Step 2

(b) If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 12$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.179$$

The rejection region then contains all values smaller than $-2.179$ and all values greater than $2.179$,thus we reject $H_0$ when $t < -2.179$ or $t > 2.179$.

Step 3

(c) Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{50 - 46 - 0}{\sqrt{\frac{5^2}{15} + \frac{15^2}{12}}} \approx 0.885$$

Step 4

(d) When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-2.179 < 0.885 < 2.179 \Rightarrow \text{ Fail to reject } H_0$$

🎯 **Result**

(a) $df = 12$

(b) Reject $H_0$ when $t < -2.179$ or $t > 2.179$

(c) $t = 0.885$

(d) Fail to reject $H_0$

**PROBLEM STATEMENT**

Chapter 11, Page 391

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 20 \\
n_2 &= \text{Sample size} = 16 \\
\overline{x}_1 &= \text{Sample mean} = 100 \\
\overline{x}_2 &= \text{Sample mean} = 94 \\
s_1 &= \text{Sample standard deviation} = 15 \\
s_2 &= \text{Sample standard deviation} = 8 \\
\alpha &= \text{Significance level} = 0.05 \\
H_0 &: \mu_1 \leq \mu_2 \\
H_1 &: \mu_1 > \mu_2
\end{aligned}
$$

(a) Determine the degrees of freedom (rounded down to the nearest integer):

$$
\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{15^2}{20} + \frac{8^2}{16}\right)^2}{\frac{(15^2/20)^2}{20 - 1} + \frac{(8^2/16)^2}{16 - 1}} \approx 30
$$

Step 2

(b) If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value $t_0$ that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 30$ and in the column with $\alpha = 0.05$ (one-tailed).

$$t = 1.697$$

The rejection region then contains all values greater than $1.697$ and thus we reject $H_0$ when the t-value is larger than 1.697.

Step 3

(c) Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{100 - 94 - 0}{\sqrt{\frac{15^2}{20} + \frac{8^2}{16}}} \approx 1.536$$

Step 4

(d) When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$1.536 < 1.697 \Rightarrow \text{ Fail to reject } H_0$$

🎯 **Result**

(a) 30

(b) Reject $H_0$ when $t > 1.697$

(c) $t = 1.536$

(d) Fail to reject $H_0$

**PROBLEM STATEMENT**

Chapter 11, Page 391

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 16 \\
n_2 &= \text{Sample size} = 18 \\
\overline{x}_1 &= \text{Sample mean} = 11045 \\
\overline{x}_2 &= \text{Sample mean} = 12840 \\
s_1 &= \text{Sample standard deviation} = 835 \\
s_2 &= \text{Sample standard deviation} = 1545 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

Given claim: Mean is larger for Russia

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 \geq \mu_2$$

$$H_1 : \mu_1 < \mu_2$$

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{835^2}{16} + \frac{1545^2}{18}\right)^2}{\frac{(835^2/16)^2}{16 - 1} + \frac{(1545^2/18)^2}{18 - 1}} \approx 26$$

Step 2

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = 26$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -1.706$$

The rejection region then contains all values smaller than $-1.706$ and thus we reject $H_0$ when the t-value is smaller than $-1.706$.

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{11045 - 12840 - 0}{\sqrt{\frac{835^2}{16} + \frac{1545^2}{18}}} \approx -4.276$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -4.276 < -1.706 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean cost is larger for adopting children from Russia.

Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 391

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 15 \\ n_2 &= \text{Sample size} = 12 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{5 + 4.5 + 3.4 + \ldots + 5}{15} = 4.3867 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{3.1 + 3.7 + 4 + \ldots + 3.6}{12} = 3.8583 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(5 - 4.3867)^2 + \ldots + (5 - 4.3867)^2}{15 - 1}} \\ &= 0.7945 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(3.1 - 3.8583)^2 + \ldots + (3.6 - 3.8583)^2}{12 - 1}} \\ &= 0.8806 \end{aligned}$$

Step 2

Given claim: The mean is more for Claiborne

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 \leq \mu_2$$

$$H_1 : \mu_1 > \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Right-tailed

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{0.7945^2}{15} + \frac{0.8806^2}{12}\right)^2}{\frac{(0.7945^2/15)^2}{15 - 1} + \frac{(0.8806^2/12)^2}{12 - 1}} \approx 22$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value $t_0$ that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 22$ and in the column with $\alpha = 0.05$.

$$t = 1.717$$

The rejection region then contains all values greater than $1.717$ and thus we reject $H_0$ when the t-value is larger than 1.717.

Step 3

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{4.3867 - 3.8583 - 0}{\sqrt{\frac{0.7945^2}{15} + \frac{0.8806^2}{12}}} \approx 1.618$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = 1.618 < 1.717 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that Claiborne models earn more.

Result

Not reasonable

**PROBLEM STATEMENT**

Chapter 11, Page 398

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 4$$
$$H_0 : \mu_d \leq 0$$
$$H_a : \mu_d > 0$$

Let us determine the difference between each pair of data values:

| Population 1 | Population 2 | Difference d |
|---|---|---|
| 10 | 8 | 10-8 = **2** |
| 12 | 9 | 12-9 = **3** |
| 15 | 12 | 15-12 = **3** |
| 19 | 15 | 19-15 = **4** |

The mean is the sum of all values divided by the number of values:

$$\overline{d} = \frac{2+3+3+4}{4} = \frac{12}{4} = 3$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s_d = \sqrt{\frac{(2-3)^2+(3-3)^2+(3-3)^2+(4-3)^2}{4-1}} \approx 0.8165$$

Step 2

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value $t_0$ that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 4 - 1 = 3$ and in the column with $\alpha = 0.05$.

$$t = 2.353$$

The rejection region then contains all values greater than $2.353$ and thus we reject $H_0$ when the t-value is larger than 2.353.

Determine the value of the test statistic:

$$t = \frac{\overline{d}}{s_d/\sqrt{n}} = \frac{3}{0.8165/\sqrt{4}} \approx 7.348$$

We reject the null hypothesis $H_0$ when the value of the test statistic is in the rejection region.

$$t = 7.348 > 2.353 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the there are more defects produced on the afternoon shift.

Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 398

**Solution A**

**Step 1**

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 5$$
$$H_0 : \mu_d = 0$$
$$H_a : \mu_d \neq 0$$

Let us determine the difference between each pair of data values:

| Population 1 | Population 2 | Difference d |
|---|---|---|
| 30 | 26 | $30 - 26 = 4$ |
| 22 | 19 | $22 - 19 = 3$ |
| 25 | 20 | $25 - 20 = 4$ |
| 19 | 15 | $19 - 15 = 4$ |
| 26 | 19 | $26 - 19 = 7$ |

The mean is the sum of all values divided by the number of values:

$$\overline{d} = \frac{4+3+4+4+7}{5} = \frac{22}{5} = 4.4$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s_d = \sqrt{\frac{(4-4.4)^2+(3-4.4)^2+(4-4.4)^2 \\ +(4-4.4)^2+(7-4.4)^2}{5-1}} \approx 1.5166$$

Step 2

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

$$\text{Two-tailed}$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 5 - 1 = 4$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.776$$

The rejection region then contains all values smaller than $-2.776$ and all values greater than $2.776$, thus we reject $H_0$ when $t < -2.776$ or $t > 2.776$.

Determine the value of the test statistic:

$$t = \frac{\overline{d}}{s_d/\sqrt{n}} = \frac{4.4}{1.5166/\sqrt{5}} \approx 6.487$$

We reject the null hypothesis $H_0$ when the value of the test statistic is in the rejection region.

$$t = 6.487 > 2.776 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is a difference in the mean number of citations given by the two officers.

Result

Yes

Solution B

Step 1

The problem wants us to test whether there is a significant difference between the mean number of citations given by Officer A and Officer B. In order to answer this, we need to conduct hypothesis testing. We need to examine the given data in order to choose an appropriate t-test for the hypothesis testing. In addition, we will follow the $6$ steps of hypothesis testing:

Step 1: Stating the hypotheses.

Step 2: Determining the level of significance.

Step 3: Determining the test statistic.

Step 4: Identifying the decision rule.

Step 5: Making a decision.

Step 6: Interpreting the results.

Step 2

## Hypothesis Testing:

**Step 1: Null and Alternate Hypotheses**

We want to know whether there is a significant difference in the mean number of citations given by Officer A and Officer B. Incorporating numbers into the problem, we let $0$ indicates that there is no difference between the population means while numbers not equal to $0$ indicates population means of the two officers differ in some other way. Based on this, our null is that difference between the two population means is equal to $0$ while our alternate hypothesis is that the difference between the two population means is not equal to $0$. And we can express these as:

$$H_0 : \mu_d = 0$$

$$H_1 : \mu_d \neq 0$$

Step 3

**Step 2: Level of Significance**

Now, we proceed in determining the level of significance. Based on the given information, we will be using $0.05$ level of significance:

$$\alpha = 0.05$$

Step 4

**Step 3: Test Statistic**

Again, we want to know whether there is a significant difference between the mean number of citations from the two officers. We treated these observations as dependent or related thus we call them **paired observations**. Based on our hypotheses, we are testing whether the difference is **equal or not equal to 0**. Since equality is involved then we will use a two-tailed test. (Remember, if inequality is involved in the hypotheses then we use a *one-tailed test*, if equality is involved then we use a *two-tailed test*). The value of test-statistic for paired samples can be calculated by:

$$t = \frac{\overline{d}}{\frac{s_d}{\sqrt{n}}}$$

where $\overline{d}$ is the mean difference, $s_d$ is the standard deviation of the differences, and $n$ is the number of observations in the sample.

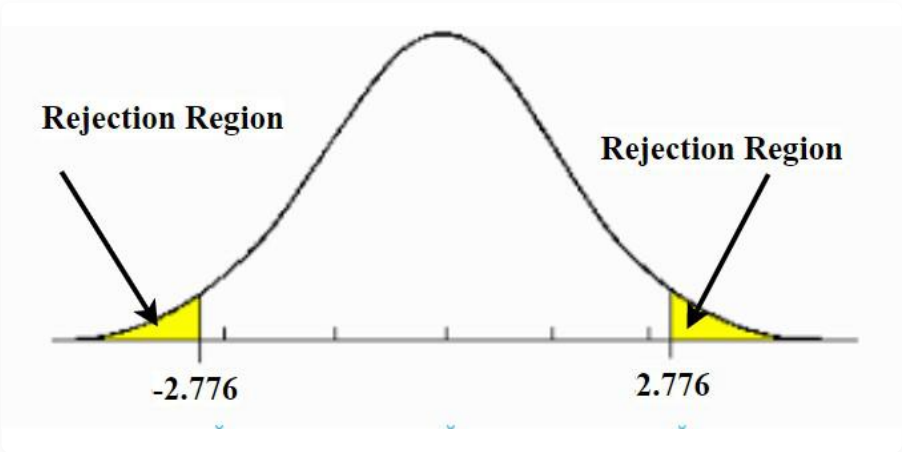
Step 5

**Step 4: Decision Rule**

To formulate the decision rule, we need to use the table for the critical values of $t$ a the two-tailed test. To locate the critical value $t$, we first determine the degrees of freedom ($df$). This value is the difference between the total number of observations and $1$. In the information given, there are $5$ paired samples. Thus,

$$df = 5 - 1 = 4$$

On the $t-$table, the first column of the table shows the degrees of freedom and the first row shows the confidence interval. The value on the column $0.05$ under the two-tailed test and row $4$ is $2.776$ but because our rejection regions are both at the end side of the distribution, then we will take the negative and positive of the value shown. Hence, our critical region is less than $-2.776$ and more than $2.776$. Illustrating we have,




Therefore, **the decision rule is to reject $H_0$ if**

$$\boldsymbol{t < -2.778} \quad \text{or} \quad \boldsymbol{t > 2.776}$$

Step 6

**Step 5: Decision**

Now that we have a decision rule, we proceed to calculating $t$. We know that our $n = 5$ for there are $5$ paired observations. We do not know the mean difference between the paired observations and the difference standard deviation yet so we need to compute for the two values using the formulas:

$$\overline{d} = \frac{\Sigma d}{n}$$

$$s_d = \sqrt{\frac{\Sigma(d - \overline{d})^2}{n - 1}}$$

Step 7

**Difference Mean:**

Taking the difference between Officer A samples and Officer B samples we have:

| Officer A | Officer B | Difference |
|---|---|---|
| $30$ | $26$ | $30 - 26 = 4$ |
| $22$ | $19$ | $22 - 19 = 3$ |
| $25$ | $20$ | $25 - 20 = 5$ |
| $19$ | $15$ | $19 - 15 = 4$ |
| $26$ | $19$ | $26 - 19 = 7$ |

Using the mean formula we obtain:

$$\begin{aligned}\overline{d} &= \frac{\Sigma d}{n} \\ &= \frac{4 + 3 + 5 + 4 + 7}{5} \\ &= \frac{23}{5} \\ &= 4.6\end{aligned}$$

Step 8

**Difference Standard Deviation:**

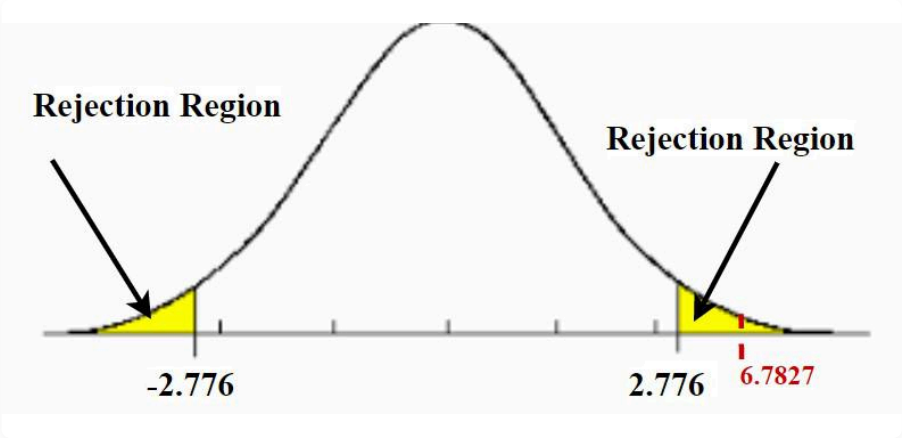
$$
\begin{aligned}
s_d &= \sqrt{\frac{\Sigma(d-\overline{d})^2}{n-1}} \\
&= \sqrt{\frac{\Sigma(4-4.6)^2+(3-4.6)^2+(5-4.6)^2+(4-4.6)^2++(7-4.6)^2}{5-1}} \\
&= \sqrt{\frac{(-0.6)^2+(-1.6)^2+(0.4)^2+(-0.6)^2+(2.4)^2}{4}} \\
&= \sqrt{\frac{0.36+2.56+0.16+0.36+5.76}{4}} \\
&= \sqrt{\frac{9.2}{4}} \\
&= \sqrt{2.3} \\
&= 1.5166
\end{aligned}
$$

Step 9

We can now proceed to calculating $t$ by substituting $\overline{d} = .6, s_d = 1.5166, n = 5$ into the $t$ formula:

$$
\begin{aligned}
t &= \frac{\overline{d}}{\frac{s_d}{\sqrt{n}}} && \text{(Write the formula.)} \\
&= \frac{4.6}{\frac{1.5166}{\sqrt{5}}} && \text{(Substitute the values.)} \\
&= \frac{4.6}{\frac{1.5166}{2.2361}} && \text{(Evaluate the square root.)} \\
&\approx 6.7827 && \text{(Simplify.)}
\end{aligned}
$$

Since the computed $t$ value is $6.7827$, then we reject the null hypothesis because $6.7827$ is greater than the critical value $2.776$ such that $6.7827$ falls into the rejection region illustrated by:




Therefore,,

$$\text{Decision: } t = 6.7827 > 2.776 \longrightarrow \text{Reject } H_0$$

Step 10

**Step 6: Interpretation**

We've just rejected the null hypothesis which says that the difference in the mean number of citations is equal to $0$. Thus, we will accept the alternate hypothesis which says that the difference in the mean number is not equal to $0$. This means that there is a significant difference between the two population means. This test will not describe their difference since we just tested whether there is a difference or not. With this, we can also say that the skills of the officers in catching over-speeding vehicles differ.

**PROBLEM STATEMENT**

Chapter 11, Page 398

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 28$$
$$\overline{d} = \text{Sample mean difference} = 42$$
$$s_d = \text{Sample standard deviation} = 2.1$$

Let us determine the difference between each pair of data values:

| **Population 1** | **Population 2** | **Difference** d |
|---|---|---|
| 320 | 340 | $320 - 340 = -20$ |
| 290 | 285 | $290 - 285 = 5$ |
| 421 | 475 | $421 - 475 = -54$ |
| 510 | 510 | $510 - 510 = 0$ |
| 210 | 210 | $210 - 210 = 0$ |
| 402 | 500 | $402 - 500 = -98$ |
| 625 | 631 | $625 - 631 = -6$ |
| 560 | 560 | $560 - 560 = 0$ |
| 360 | 365 | $360 - 365 = -5$ |
| 431 | 431 | $431 - 431 = 0$ |
| 506 | 525 | $506 - 525 = -19$ |
| 505 | 619 | $505 - 619 = -114$ |

The mean is the sum of all values divided by the number of values:

$$\overline{d} = \frac{\begin{array}{c}-20+5-54+0+0-98\\-6+0-5+0-19-114\end{array}}{12} = \frac{-311}{12} = -25.9167$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s_d = \sqrt{\frac{(-20-(-25.9167))^2 + .... + (-114-(-25.9167))^2}{12-1}} \approx 40.7909$$

Step 2

Given claim: Mean is negative (increase)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu_d \geq 0$$
$$H_a : \mu_d < 0$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 12 - 1 = 11$ and in the column with $\alpha = 0.05$.

$$t_0 = 1.796$$

The rejection region then contains all values smaller than $-1.796$ and thus we reject $H_0$ when the t-value is less than $-1.796$.

Determine the value of the test statistic:

$$t = \frac{\overline{d}}{s_d/\sqrt{n}} = \frac{-25.9167}{40.7909/\sqrt{12}} \approx -2.201$$

We reject the null hypothesis $H_0$ when the value of the test statistic is in the rejection region.

$$t = -2.201 < -1.796 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is a significant increase in the typical salesperson's weekly income due to the innovative incentive plan.

Step 3

### P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value $2.201$ in the row $df = n - 1 = 12 - 1 = 11$ (one-tailed):

$$P = 0.05$$

🎯 Result

Yes, $P = 0.05$

**PROBLEM STATEMENT**

Chapter 11, Page 399

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 8$$

Given claim: Mean is positive (decrease)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu_d \leq 0$$
$$H_a : \mu_d > 0$$

Let us determine the difference between each pair of data values:

| **Population 1** | **Population 2** | **Difference** d |
|---|---|---|
| 14 | 2 | $14 - 2 = 12$ |
| 7 | 7 | $7 - 7 = 0$ |
| 4 | 3 | $4 - 3 = 1$ |
| 5 | 6 | $5 - 6 = -1$ |
| 17 | 8 | $17 - 8 = 9$ |
| 12 | 13 | $12 - 13 = -1$ |
| 8 | 3 | $8 - 3 = 5$ |
| 9 | 5 | $9 - 5 = 4$ |

The mean is the sum of all values divided by the number of values:

$$\overline{d} = \frac{\begin{array}{c}12 + 0 + 1 - 1 \\ +9 - 1 + 5 + 4\end{array}}{8} = \frac{29}{8} = 3.625$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_d = \sqrt{\frac{(12 - 3.625)^2 + \ldots + (4 - 3.625)^2}{8 - 1}} \approx 4.8385$$

Step 2

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value $t_0$ that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 8 - 1 = 7$ and in the column with $\alpha = 0.05$.

$$t = 1.895$$

The rejection region then contains all values greater than $1.895$ and thus we reject $H_0$ when the t-value is larger than 1.895.

Determine the value of the test statistic:

$$t = \frac{\overline{d}}{s_d/\sqrt{n}} = \frac{3.625}{4.8385/\sqrt{8}} \approx 2.119$$

We reject the null hypothesis $H_0$ when the value of the test statistic is in the rejection region.

$$t = 2.119 > 1.895 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that there is a decrease in the number of crimes since the inauguration of the program.

Step 3

### P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value $2.119$ in the row $df = n - 1 = 8 - 1 = 7$ (one-tailed):

$$0.025 < P < 0.05$$

Note: When using technology, then you obtain the P-value 0.03591.

🎯 **Result**

Yes, $0.025 < P < 0.05$

**PROBLEM STATEMENT**

Chapter 11, Page 400

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 35 \\
n_2 &= \text{Sample size} = 40 \\
\overline{x}_1 &= \text{Sample mean} = 24.51 \\
\overline{x}_2 &= \text{Sample mean} = 22.69 \\
\sigma_1 &= \text{Population standard deviation} = 4.48 \\
\sigma_2 &= \text{Population standard deviation} = 3.86 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$
\begin{aligned}
H_0 &: \mu_1 = \mu_2 \\
H_a &: \mu_1 \neq \mu_2
\end{aligned}
$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Two-tailed test

The rejection region of a two-tailed test with $\alpha = 0.01$ contains all z-values below the z-value $-z_0$ that has a probability of $0.01/2 = 0.005$ to its left and all z-values above the z-value $z_0$ that has a probability of $0.01/2 = 0.005$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.01}{2} = 0.005$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.005 = 0.495$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.495 is closest to 0.4951, which is given in the row starting with "2.5" and in the column starting with ".08", thus the corresponding z-score is $2.5 + .08 = 2.58$.

$$z_0 = \pm 2.58$$

Thus the rejection region then contains all z-scores smaller than $-2.58$ and all z-scores larger than 2.58, thus we reject $H_0$ when the z-score is smaller than $-2.58$ or the z-score is larger than 2.58.

Determine the value of the test statistic:

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{24.51 - 22.69 - 0}{\sqrt{\frac{4.48^2}{35} + \frac{3.86^2}{40}}} \approx 1.87$$

If the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$.

$$-2.58 < 1.87 < 2.58 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that there is a difference in the mean number of times men and women order take-out dinners in a month.

Step 3

### P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic $z$, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -1.87$ or larger than $z = 1.87$.

$$P(z < -1.87 \text{ or } z > 1.87)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.87)$ is given in the row starting with "$1.8$" and in the column starting with ".07" of the table.

$$\begin{aligned} P &= P(z < -2.29 \text{ or } z > 2.29) \\ &= 2P(z > 2.29) \\ &= 2(1 - P(z < 2.29)) \\ &= 2(1 - P(z < 0) - P(0 < z < 2.29)) \\ &= 2(1 - 0.5000 - 0.4693) \\ &= 2(0.0307) \\ &= 0.0614 \\ &= 6.14\% \end{aligned}$$

Result

Yes, $P = 0.0614$

**PROBLEM STATEMENT**

Chapter 11, Page 400

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 54 \\
n_2 &= \text{Sample size} = 60 \\
\overline{x}_1 &= \text{Sample mean} = 345 \\
\overline{x}_2 &= \text{Sample mean} = 351 \\
\sigma_1 &= \text{Population standard deviation} = 21 \\
\sigma_2 &= \text{Population standard deviation} = 28 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

Given claim: Mean is larger for night shift

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$
\begin{aligned}
H_0 &: \mu_1 \geq \mu_2 \\
H_a &: \mu_1 < \mu_2
\end{aligned}
$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Left-tailed test

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all z-scores below the z-score $z_0$ that has a probability of 0.05 to its left.

$$P(z < z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = -1.65$$

Thus the rejection region then contains all z-scores smaller than $-1.65$ and thus we reject $H_0$ when the z-score is smaller than $-1.65$.

Determine the value of the test statistic:

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{345 - 351 - 0}{\sqrt{\frac{21^2}{54} + \frac{28^2}{60}}} \approx -1.30$$

If the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$.

$$-1.30 > -1.65 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the number of units produced on the night shift larger.

Result

No

**PROBLEM STATEMENT**

Chapter 11, Page 400

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 40 \\
n_2 &= \text{Sample size} = 50 \\
\overline{x}_1 &= \text{Sample mean} = 4.77 \\
\overline{x}_2 &= \text{Sample mean} = 5.02 \\
\sigma_1 &= \text{Population standard deviation} = 1.05 \\
\sigma_2 &= \text{Population standard deviation} = 1.23 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

Given claim: Difference in the means

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$
\begin{aligned}
H_0 &: \mu_1 = \mu_2 \\
H_a &: \mu_1 \neq \mu_2
\end{aligned}
$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Two-tailed test

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all z-values below the z-value $-z_0$ that has a probability of $0.05/2 = 0.025$ to its left and all z-values above the z-value $z_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.025 = 0.475$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.475 is closest to 0.4750, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = \pm 1.96$$

Thus the rejection region then contains all z-scores smaller than $-1.96$ and all z-scores larger than 1.96, thus we reject $H_0$ when the z-score is smaller than $-1.96$ or the z-score is larger than 1.96.

Determine the value of the test statistic:

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{4.77 - 5.02 - 0}{\sqrt{\frac{1.05^2}{40} + \frac{1.23^2}{50}}} \approx -1.04$$

If the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$.

$$-1.96 < -1.04 < 1.96 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that there is a difference in the mean number of calls per day between the two employees.

Step 3

### P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic $z$, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -1.04$ or larger than $z = 1.04$.

$$P(z < -1.04 \text{ or } z > 1.04)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.04)$ is given in the row starting with "1.0" and in the column starting with ".04" of the table.

$$\begin{aligned} P &= P(z < -1.04 \text{ or } z > 1.04) \\ &= 2P(z > 1.04) \\ &= 2(1 - P(z < 1.04)) \\ &= 2(1 - P(z < 0) - P(0 < z < 1.04)) \\ &= 2(1 - 0.5000 - 0.3508) \\ &= 2(0.1492) \\ &= 0.2984 \\ &= 29.84\% \end{aligned}$$

Result

No, $P = 0.2984$

**PROBLEM STATEMENT**

Chapter 11, Page 401

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 50 \\
n_2 &= \text{Sample size} = 40 \\
\overline{x}_1 &= \text{Sample mean} = 4.35 \\
\overline{x}_2 &= \text{Sample mean} = 5.84 \\
\sigma_1 &= \text{Population standard deviation} = 1.20 \\
\sigma_2 &= \text{Population standard deviation} = 1.36 \\
\alpha &= \text{Significance level} = 0.01
\end{aligned}
$$

Given claim: Mean is less for the regular-coffee drinkers

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$
\begin{aligned}
H_0 &: \mu_1 \geq \mu_2 \\
H_a &: \mu_1 < \mu_2
\end{aligned}
$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Left-tailed test

The rejection region of a left-tailed test with $\alpha = 0.01$ contains all z-scores below the z-score $z_0$ that has a probability of 0.01 to its left.

$$P(z < z_0) = 0.01$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.01 = 0.49$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.49 is closest to 0.4901, which is given in the row starting with "2.3" and in the column starting with ".03", thus the corresponding z-score is $2.3 + .03 = 2.33$.

$$z_0 = -2.33$$

Thus the rejection region then contains all z-scores smaller than $-2.33$ and thus we reject $H_0$ when the z-score is smaller than $-2.33$.

Determine the value of the test statistic:

$$z = \frac{\overline{x}_1 - \overline{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{4.35 - 5.84 - 0}{\sqrt{\frac{1.20^2}{50} + \frac{1.36^2}{40}}} \approx -5.44$$

If the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$.

$$-5.44 < -2.33 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean daily consumption of regular-coffee drinkers is less than that of decaffeinated-coffee drinkers.

Step 3

### P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic $z$, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -5.44$ or larger than $z = 5.44$.

$$P(z < -5.44 \text{ or } z > 5.44)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 5.44)$ is given in the row starting with "$5.4$" and in the column starting with ".04" of the table. Since the table contains no row starting with "5.4", we estimate the probability as 0.5000.

$$\begin{aligned} P &= P(z < -5.44 \text{ or } z > 5.44) \\ &= 2P(z > 5.44) \\ &= 2(1 - P(z < 5.44)) \\ &= 2(1 - P(z < 0) - P(0 < z < 5.44)) \\ &\approx 2(1 - 0.5000 - 0.5000) \\ &= 2(0) \\ &= 0 \\ &= 0\% \end{aligned}$$

Result

There is sufficient evidence to support the claim that the mean daily consumption of regular-coffee drinkers is less than that of decaffeinated-coffee drinkers.

$$P = 0$$

## Chapter 11 Exercise 31

Step 1

4. Step & Result Cards Not found

**PROBLEM STATEMENT**

Chapter 11, Page 401

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 40 \\
n_2 &= \text{Sample size} = 30 \\
\overline{x}_1 &= \text{Sample mean} = 57000 \\
\overline{x}_2 &= \text{Sample mean} = 61000 \\
s_1 &= \text{Sample standard deviation} = 9200 \\
s_2 &= \text{Sample standard deviation} = 7100 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

Given claim: Mean is larger for plan B

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 \geq \mu_2$$

$$H_1 : \mu_1 < \mu_2$$

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{9200^2}{40} + \frac{7100^2}{30}\right)^2}{\frac{(9200^2/40)^2}{40 - 1} + \frac{(7100^2/30)^2}{30 - 1}} \approx 67$$

**Step 2**

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.10$ contains all t-value below the t-value $-t_0$ that has a probability of 0.05 to its left.

$$P(t < -t_0) = 0.05$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = 67$ and in the column with $\alpha = 0.05$ (which is the negation of the critical value in the table).

$$t = -1.668$$

The rejection region then contains all values smaller than $-1.668$ and thus we reject $H_0$ when the t-value is smaller than $-1.668$.

**Step 3**

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{57000 - 61000 - 0}{\sqrt{\frac{9200^2}{40} + \frac{7100^2}{30}}} \approx -2.053$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -2.053 < -1.668 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the mean income of those selecting Plan B is larger.

Step 4

### P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value $2.053$ in the row $df = 67$ (one-tailed):

$$0.01 < P < 0.025$$

Note: When using technology, then you obtain the P-value 0.02199. The coefficient of skewness do n

🎯 Result

Yes, $0.01 < P < 0.025$, No

**PROBLEM STATEMENT**

Chapter 11, Page 401

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 35 \\
n_2 &= \text{Sample size} = 45 \\
\overline{x}_1 &= \text{Sample mean} = 18 \\
\overline{x}_2 &= \text{Sample mean} = 15.5 \\
s_1 &= \text{Sample standard deviation} = 4.2 \\
s_2 &= \text{Sample standard deviation} = 3.9 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

Given claim: The mean is larger for software

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 \leq \mu_2$$

$$H_1 : \mu_1 > \mu_2$$

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{4.2^2}{35} + \frac{3.9^2}{45}\right)^2}{\frac{(4.2^2/35)^2}{35 - 1} + \frac{(3.9^2/45)^2}{45 - 1}} \approx 70$$

**Step 2**

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value $t_0$ that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 70$ and in the column with $\alpha = 0.05$.

$$t = 1.667$$

The rejection region then contains all values greater than $1.667$ and thus we reject $H_0$ when the t-value is larger than 1.667.

**Step 3**

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{18 - 15.5 - 0}{\sqrt{\frac{4.2^2}{35} + \frac{3.9^2}{45}}} \approx 2.724$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = 2.724 > 1.667 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that it does take longer to resolve software issues.

Step 4

### P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value $2.724$ in the row $df = 70$ (one-tailed):

$$0.0005 < P < 0.05$$

Note: When using technology, then you obtain the P-value $0.00407$.

🎯 Result

Yes, $0.0005 < P < 0.05$

**PROBLEM STATEMENT**

Chapter 11, Page 401

Step 1

Given:

$$n_1 = \text{Sample size} = 200$$
$$n_2 = \text{Sample size} = 300$$
$$x_1 = \text{Number of successes} = 180$$
$$x_2 = \text{Number of successes} = 261$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Proportion of Advil is lower (more effective)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi_1 \leq \pi_2$$
$$H_a : \pi_1 > \pi_2$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score $z_0$ that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than $1.65$ and thus we reject $H_0$ when the z-score is greater than 1.65.

Step 3

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{180}{200} \approx 0.9$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{261}{300} \approx 0.87$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{180 + 261}{200 + 300} = \frac{441}{500} \approx 0.882$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.9 - 0.87}{\sqrt{0.882(1 - 0.882)}\sqrt{\frac{1}{200} + \frac{1}{300}}} \approx 1.02$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$z = 1.02 < 1.65 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the new drug is more effective.

🎯 Result

No

**PROBLEM STATEMENT**

Chapter 11, Page 401

Step 1

Given:

$$n_1 = \text{Sample size} = 300$$
$$n_2 = \text{Sample size} = 290$$
$$x_1 = \text{Number of successes} = 160$$
$$x_2 = \text{Number of successes} = 170$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Proportion is larger for this month

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi_1 \geq \pi_2$$
$$H_a : \pi_1 < \pi_2$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Left-tailed test

The rejection region of a left-tailed test with $\alpha = 0.05$ contains all z-scores below the z-score $z_0$ that has a probability of 0.05 to its left.

$$P(z < z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = -1.65$$

Thus the rejection region then contains all z-scores smaller than $-1.65$ and thus we reject $H_0$ when the z-score is smaller than $-1.65$.

Step 3

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{160}{300} \approx 0.5333$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{170}{290} \approx 0.5862$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{160 + 170}{300 + 290} = \frac{330}{590} \approx 0.5593$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.5333 - 0.5862}{\sqrt{0.5593(1 - 0.5593)}\sqrt{\frac{1}{300} + \frac{1}{290}}} \approx -1.29$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$z = -1.29 > -1.65 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that a larger proportion of the agents believe the economy is contracting this month.

🎯 Result

No

**PROBLEM STATEMENT**

Chapter 11, Page 401

Step 1

Given:

$$n_1 = \text{Sample size} = 1500$$
$$n_2 = \text{Sample size} = 1600$$
$$x_1 = \text{Number of successes} = 990$$
$$x_2 = \text{Number of successes} = 970$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: Proportion is larger for men

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi_1 \leq \pi_2$$
$$H_a : \pi_1 > \pi_2$$

**Step 2**

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all z-scores above the z-score $z_0$ that has a probability of 0.01 to its right.

$$P(z > z_0) = 0.01$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.01 = 0.49$ in the normal probability table of the appendix. We note that the probability 0.49 is closest to 0.4901, which is given in the row starting with "2.3" and in the column starting with ".03", thus the corresponding z-score is $2.3 + .03 = 2.33$.

$$z_0 = 2.33$$

Thus the rejection region then contains all z-scores greater than $2.33$ and thus we reject $H_0$ when the z-score is greater than 2.33.

**Step 3**

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{990}{1500} \approx 0.66$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{970}{1600} \approx 0.60625$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{990 + 970}{1500 + 1600} = \frac{1960}{3100} \approx 0.6323$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.66 - 0.60625}{\sqrt{0.6323(1 - 0.6323)}\sqrt{\frac{1}{1500} + \frac{1}{1600}}} \approx 3.10$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$z = 3.10 < 2.33 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the proportion of men who believe the division of household duties is fair is larger.

Step 4

### P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic $z$, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -3.10$ or larger than $z = 3.10$.

$$P(z < -3.10 \text{ or } z > 3.10)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 3.10)$ is given in the row starting with "$3.1$" and in the column starting with ".00" of the table.

$$\begin{aligned} P &= P(z < -3.10 \text{ or } z > 3.10) \\ &= 2P(z > 3.10) \\ &= 2(1 - P(z < 3.10)) \\ &= 2(1 - P(z < 0) - P(0 < z < 3.10)) \\ &= 2(1 - 0.5000 - 0.4990) \\ &= 2(0.0010) \\ &= 0.0020 \\ &= 0.20\% \end{aligned}$$

Result

Yes, $P = 0.0020$

**PROBLEM STATEMENT**

Chapter 11, Page 401

Step 1

Given:

$$n_1 = \text{Sample size} = 500$$
$$n_2 = \text{Sample size} = 400$$
$$x_1 = \text{Number of successes} = 450$$
$$x_2 = \text{Number of successes} = 352$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: Difference in the proportions

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi_1 = \pi_2$$
$$H_a : \pi_1 \neq \pi_2$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Two-tailed test

The rejection region of a two-tailed test with $\alpha = 0.01$ contains all z-values below the z-value $-z_0$ that has a probability of $0.01/2 = 0.005$ to its left and all z-values above the z-value $z_0$ that has a probability of $0.01/2 = 0.005$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.01}{2} = 0.005$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.005 = 0.495$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.495 is closest to 0.4951, which is given in the row starting with "2.5" and in the column starting with ".08", thus the corresponding z-score is $2.5 + .08 = 2.58$.

$$z_0 = \pm 2.58$$

Thus the rejection region then contains all z-scores smaller than $-2.58$ and all z-scores larger than 2.58, thus we reject $H_0$ when the z-score is smaller than $-2.58$ or the z-score is larger than 2.58.

Step 3

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{450}{500} \approx 0.9$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{352}{400} \approx 0.88$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{450 + 352}{500 + 400} = \frac{802}{900} \approx 0.8911$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.9 - 0.88}{\sqrt{0.8911(1 - 0.8911)}\sqrt{\frac{1}{500} + \frac{1}{400}}} \approx 0.96$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-2.58 < 0.96 < 2.58 \Rightarrow \text{ Fail to reject } H_0$$

There is sufficient evidence to support the claim that there is a difference in the percent of time that access to the Internet is successful.

🎯 Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 401

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 5000 \\
n_2 &= \text{Sample size} = 5000 \\
\hat{p}_1 &= \text{Sample proportion} = 9.1\% = 0.091 \\
\hat{p}_2 &= \text{Sample proportion} = 8.5\% = 0.085 \\
x_1 &= \text{Number of successes} = n\hat{p}_1 = 5000(0.091) = 455 \\
x_2 &= \text{Number of successes} = n\hat{p}_1 = 5000(0.085) = 425 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

Given claim: Proportion is higher last month (increase)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$
\begin{aligned}
H_0 &: \pi_1 \leq \pi_2 \\
H_a &: \pi_1 > \pi_2
\end{aligned}
$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score $z_0$ that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than $1.65$ and thus we reject $H_0$ when the z-score is greater than 1.65.

Step 3

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{455 + 425}{5000 + 5000} = \frac{880}{10000} \approx 0.0880$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.091 - 0.085}{\sqrt{0.0880(1 - 0.0880)}\sqrt{\frac{1}{5000} + \frac{1}{5000}}} \approx 1.06$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$z = 1.06 < 1.65 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that there is an increase in the share asserting conditions are "good".

Step 4

### P-value

The P-value is the probability of obtaining a value more extreme or equal to the standardized test statistic $z$, assuming that the null hypothesis is true.

Since the test is two-tailed, the P-value is the probability that the z-score is smaller than $z = -1.06$ or larger than $z = 1.06$.

$$P(z < -1.06 \text{ or } z > 1.06)$$

Determine corresponding probability using the normal probability table in the appendix. The probability $P(0 < z < 1.06)$ is given in the row starting with "$1.0$" and in the column starting with ".06" of the table.

$$\begin{aligned} P &= P(z < -1.06 \text{ or } z > 1.06) \\ &= 2P(z > 1.06) \\ &= 2(1 - P(z < 1.06)) \\ &= 2(1 - P(z < 0) - P(0 < z < 1.06)) \\ &= 2(1 - 0.5000 - 0.3554) \\ &= 2(0.1446) \\ &= 0.2892 \\ &= 28.92\% \end{aligned}$$

🎯 Result

There is not sufficient evidence to support the claim that there is an increase in the share asserting conditions are "good".

$$P = 0.2892$$

**PROBLEM STATEMENT**

Chapter 11, Page 402

Step 1

Given:

$$n_1 = \text{Sample size} = 270$$
$$n_2 = \text{Sample size} = 203$$
$$x_1 = \text{Number of successes} = 56$$
$$x_2 = \text{Number of successes} = 52$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Difference in the proportions

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi_1 = \pi_2$$
$$H_a : \pi_1 \neq \pi_2$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Two-tailed test

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all z-values below the z-value $-z_0$ that has a probability of $0.05/2 = 0.025$ to its left and all z-values above the z-value $z_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.025 = 0.475$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.475 is closest to 0.4750, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = \pm 1.96$$

Thus the rejection region then contains all z-scores smaller than $-1.96$ and all z-scores larger than 1.96, thus we reject $H_0$ when the z-score is smaller than $-1.96$ or the z-score is larger than 1.96.

Step 3

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{56}{270} \approx 0.2074$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{52}{203} \approx 0.2562$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{56 + 52}{270 + 203} = \frac{108}{473} \approx 0.2283$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.2074 - 0.2562}{\sqrt{0.2283(1 - 0.2283)}\sqrt{\frac{1}{270} + \frac{1}{203}}} \approx -1.25$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-1.96 < -1.25 < 1.96 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that there is a difference in the proportion of humorous ads in British versus american trade magazines.

🎯 Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 402

Step 1

Given:

$$n_1 = \text{Sample size} = 300$$
$$n_2 = \text{Sample size} = 200$$
$$x_1 = \text{Number of successes} = 100$$
$$x_2 = \text{Number of successes} = 36$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Difference in the proportions

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi_1 = \pi_2$$
$$H_a : \pi_1 \neq \pi_2$$

Step 2

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Two-tailed test

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all z-values below the z-value $-z_0$ that has a probability of $0.05/2 = 0.025$ to its left and all z-values above the z-value $z_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.025 = 0.475$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.475 is closest to 0.4750, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = \pm 1.96$$

Thus the rejection region then contains all z-scores smaller than $-1.96$ and all z-scores larger than 1.96, thus we reject $H_0$ when the z-score is smaller than $-1.96$ or the z-score is larger than 1.96.

Step 3

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{100}{300} \approx 0.3333$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{36}{200} \approx 0.18$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{100 + 36}{300 + 200} = \frac{136}{500} \approx 0.272$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.3333 - 0.18}{\sqrt{0.272(1 - 0.272)}\sqrt{\frac{1}{300} + \frac{1}{200}}} \approx 3.77$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$z = 3.77 > 1.96 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference between the responses of women and men.

🎯 Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 402

Step 1

Given:

$$n_1 = \text{Sample size} = 388$$
$$n_2 = \text{Sample size} = 307$$
$$x_1 = \text{Number of successes} = 39$$
$$x_2 = \text{Number of successes} = 11$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Proportion is higher for NBA

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi_1 \leq \pi_2$$
$$H_a : \pi_1 > \pi_2$$

**Step 2**

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Right-tailed test

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all z-scores above the z-score $z_0$ that has a probability of 0.05 to its right.

$$P(z > z_0) = 0.05$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.05 = 0.45$ in the normal probability table of the appendix. We note that the probability 0.45 is closest to 0.4505, which is given in the row starting with "1.6" and in the column starting with ".05", thus the corresponding z-score is $1.6 + .05 = 1.65$.

$$z_0 = 1.65$$

Thus the rejection region then contains all z-scores greater than $1.65$ and thus we reject $H_0$ when the z-score is greater than 1.65.

**Step 3**

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{39}{388} \approx 0.1005$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{11}{307} \approx 0.0358$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{39 + 11}{388 + 307} = \frac{50}{695} \approx 0.0719$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.1005 - 0.0358}{\sqrt{0.0719(1 - 0.0719)}\sqrt{\frac{1}{388} + \frac{1}{307}}} \approx 3.28$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$z = 3.28 > 1.65 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the NBA has more black participation in higher levels of management.

🎯 Result

There is sufficient evidence to support the claim that the NBA has more black participation in higher levels of management.

**PROBLEM STATEMENT**

Chapter 11, Page 402

Step 1

Given:

$$
\begin{aligned}
n_1 &= \text{Sample size} = 12 \\
n_2 &= \text{Sample size} = 12 \\
\overline{x}_1 &= \text{Sample mean} = 1.65 \\
\overline{x}_2 &= \text{Sample mean} = 2.2 \\
s_1 &= \text{Sample standard deviation} = 0.56 \\
s_2 &= \text{Sample standard deviation} = 0.30 \\
\alpha &= \text{Significance level} = 0.05
\end{aligned}
$$

Given claim: No difference in the means

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

Step 2

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{0.56^2}{12} + \frac{0.30^2}{12}\right)^2}{\frac{(0.56^2/12)^2}{12 - 1} + \frac{(0.30^2/12)^2}{12 - 1}} \approx 16$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 16$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.120$$

The rejection region then contains all values smaller than $-2.120$ and all values greater than $2.120$,thus we reject $H_0$ when $t < -2.120$ or $t > 2.120$.

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{1.65 - 2.2 - 0}{\sqrt{\frac{0.56^2}{12} + \frac{0.30^2}{12}}} \approx -2.999$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -2.999 < -2.120 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to reject the claim of no difference in the mean number of households picking either variety of service to download songs.

🎯 **Result**

There is sufficient evidence to reject the claim of no difference in the mean number of households picking either variety of service to download songs.

**PROBLEM STATEMENT**

Chapter 11, Page 402

Step 1

Given:

$$n_1 = \text{Sample size} = 35$$
$$n_2 = \text{Sample size} = 25$$
$$\overline{x}_1 = \text{Sample mean} = 33.6$$
$$\overline{x}_2 = \text{Sample mean} = 39.5$$
$$s_1 = \text{Sample standard deviation} = 10.9$$
$$s_2 = \text{Sample standard deviation} = 4.8$$
$$\alpha = \text{Significance level} = 0.01$$

Given claim: No difference in the means

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

**Step 2**

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{10.9^2}{35} + \frac{4.8^2}{25}\right)^2}{\frac{(10.9^2/35)^2}{35 - 1} + \frac{(4.8^2/25)^2}{25 - 1}} \approx 49$$

The rejection region of a two-tailed test with $\alpha = 0.01$ contains all t-values below the t-value $-t_0$ that has a probability of $0.01/2 = 0.005$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.01/2 = 0.005$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.01}{2} = 0.005$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 49$ and in the column with $\alpha = 0.01$ (two-tailed).

$$t = 2.680$$

The rejection region then contains all values smaller than $-2.680$ and all values greater than $2.680$,thus we reject $H_0$ when $t < -2.680$ or $t > 2.680$.

**Step 3**

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{33.6 - 39.5 - 0}{\sqrt{\frac{10.9^2}{35} + \frac{4.8^2}{25}}} \approx -2.840$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -2.840 < -2.680 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to reject the claim of no difference in the mean ages of the two groups of coupon clients.

🎯 **Result**

There is sufficient evidence to reject the claim of no difference in the mean ages of the two groups of coupon clients.

**PROBLEM STATEMENT**

Chapter 11, Page 402

Step 1

Given:

$$n_1 = \text{Sample size} = 10$$
$$n_2 = \text{Sample size} = 12$$
$$\overline{x}_1 = \text{Sample mean} = 83.55$$
$$\overline{x}_2 = \text{Sample mean} = 78.80$$
$$s_1 = \text{Sample standard deviation} = 10.50$$
$$s_2 = \text{Sample standard deviation} = 14.25$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

Step 2

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 10 + 12 - 2 = 20$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.086$$

The rejection region then contains all values smaller than $-2.086$ and all values greater than $2.086$,thus we reject $H_0$ when $t < -2.086$ or $t > 2.086$.

Step 3

Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(10 - 1)10.50^2 + (12 - 1)14.25^2}{10 + 12 - 2}} \approx 12.7003$$

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{83.55 - 78.80}{12.7003\sqrt{\frac{1}{10} + \frac{1}{12}}} \approx 0.873$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-2.086 < 0.873 < 2.086 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean number of hamburgers sold at the two locations.

Step 4

### P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value $0.873$ in the row $df = n_1 + n_2 - 2 = 10 + 12 - 2 = 20$ (two-tailed):

$$P > 0.20$$

Note: When using technology, then you obtain the P-value $0.39302$.

Result

No, $P > 0.20$

**PROBLEM STATEMENT**

Chapter 11, Page 402

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 10 \\ n_2 &= \text{Sample size} = 14 \\ \alpha &= \text{Significance level} = 0.10 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{86 + 78 + 66 + \ldots + 102}{10} = 83.8 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{91 + 71 + 75 + \ldots + 72}{14} = 79.2857 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(86 - 83.8)^2 + \ldots + (102 - 83.8)^2}{10 - 1}} \\ &= 13.6772 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(91 - 79.2857)^2 + \ldots. + (72 - 79.2857)^2}{14 - 1}} \\ &= 6.7074 \end{aligned}$$

Step 2

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{13.6772^2}{10} + \frac{6.4634^2}{14}\right)^2}{\frac{(13.6772^2/10)^2}{10 - 1} + \frac{(6.4634^2/14)^2}{14 - 1}} \approx 11$$

The rejection region of a two-tailed test with $\alpha = 0.10$ contains all t-values below the t-value $-t_0$ that has a probability of $0.10/2 = 0.05$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.10/2 = 0.05$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.10}{2} = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 11$ and in the column with $\alpha = 0.10$ (two-tailed).

$$t = 1.796$$

The rejection region then contains all values smaller than $-1.796$ and all values greater than $1.796$,thus we reject $H_0$ when $t < -1.796$ or $t > 1.796$.

Step 3

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{83.8 - 79.2857 - 0}{\sqrt{\frac{13.6772^2}{10} + \frac{6.7074^2}{14}}} \approx 0.964$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-1.796 < 0.964 < 1.796 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the length of times the two types of treatment lasted.

Result

There is not sufficient evidence to support the claim of a difference in the length of times the two types of treatment lasted.

**PROBLEM STATEMENT**

Chapter 11, Page 402

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 10 \\ n_2 &= \text{Sample size} = 14 \\ \alpha &= \text{Significance level} = 0.01 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{17.58 + 19.73 + 12.61 + \ldots + 15.85}{10} = 15.8690 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{18.19 + 20.22 + 17.38 + \ldots + 14.83}{14} = 18.2921 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n-1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(17.58 - 15.8690)^2 + \ldots + (15.85 - 15.8690)^2}{10-1}} \\ &= 2.3287 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(18.19 - 18.2921)^2 + \ldots + (14.83 - 18.2921)^2}{14-1}} \\ &= 2.5527 \end{aligned}$$

Step 2

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.01/2 = 0.005$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.01/2 = 0.005$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.01}{2} = 0.005$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 10 + 14 - 2 = 22$ and in the column with $\alpha = 0.01$ (two-tailed).

$$t = 2.819$$

The rejection region then contains all values smaller than $-2.819$ and all values greater than $2.819$,thus we reject $H_0$ when $t < -2.819$ or $t > 2.819$.

Step 3

Determine the pooled standard deviation:

$$\begin{aligned} s_p &= \sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}} \\ &= \sqrt{\frac{(10-1)2.3287^2+(14-1)2.5527^2}{10+14-2}} \\ &\approx 2.4635 \end{aligned}$$

Determine the test statistic:

$$t = \frac{\overline{x}_1-\overline{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{15.8690-18.2921}{2.4635\sqrt{\frac{1}{10}+\frac{1}{14}}} \approx -2.376$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-2.819 < 2.376 < 2.819 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean amounts purchased on impulse at the two stores.

Result

No

**PROBLEM STATEMENT**

Chapter 11, Page 402

Step 1

Given:

$$n_1 = \text{Sample size} = 12$$
$$n_2 = \text{Sample size} = 12$$
$$\alpha = \text{Significance level} = 0.05$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\overline{x}_1 = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{31.73 + 28.77 + ... + 26.49}{10} = 27.4610$$
$$\overline{x}_2 = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{22.93 + 23.92 + ... + 22.31}{12} = 25.6892$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum(x - \overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(31.73 - 27.4610)^2 + ... + (26.49 - 27.4610)^2}{10-1}} \\ &= 4.4404 \end{aligned}$$

$$\begin{aligned} s_2 &= \sqrt{\frac{\sum(x - \overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(22.93 - 25.6892)^2 + .... + (22.31 - 25.6892)^2}{12-1}} \\ &= 2.6845 \end{aligned}$$

Step 2

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{4.4404^2}{10} + \frac{2.6845^2}{12}\right)^2}{\frac{(4.4404^2/10)^2}{10 - 1} + \frac{(2.6845^2/12)^2}{12 - 1}} \approx 14$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 14$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.145$$

The rejection region then contains all values smaller than $-2.145$ and all values greater than $2.145$,thus we reject $H_0$ when $t < -2.145$ or $t > 2.145$.

Step 3

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{27.4610 - 25.6892 - 0}{\sqrt{\frac{4.4404^2}{10} + \frac{2.6845^2}{12}}} \approx 1.105$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-2.145 < 1.105 < 2.145 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean waiting time.

Result

No

**PROBLEM STATEMENT**

Chapter 11, Page 403

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 8 \\ n_2 &= \text{Sample size} = 11 \\ \alpha &= \text{Significance level} = 0.01 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{10+10+11+...+9}{8} = 10.3750 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{4+8+7+...+5}{11} = 5.6364 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n-1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum(x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(10-10.3750)^2+....+(9-10.3750)^2}{8-1}} \\ &= 2.2638 \\ s_2 &= \sqrt{\frac{\sum(x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(4-5.6364)^2+...+(5-5.6364)^2}{11-1}} \\ &= 2.4606 \end{aligned}$$

Step 2

Given claim: Mean is higher for under 25 (younger customers)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 \leq \mu_2$$

$$H_1 : \mu_1 > \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value $t_0$ that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 8 + 11 - 2 = 17$ and in the column with $\alpha = 0.01$.

$$t = 2.567$$

The rejection region then contains all values greater than $2.567$ and thus we reject $H_0$ when the t-value is larger than 2.567.

Step 3

Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(8 - 1)2.2638^2 + (11 - 1)2.4606^2}{8 + 11 - 2}} \approx 2.3815$$

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{10.3750 - 5.6364}{2.3815\sqrt{\frac{1}{8} + \frac{1}{11}}} \approx 4.282$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = 4.282 > 2.567 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that younger customers us the ATMs more.

🎯 Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 403

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 10 \\ n_2 &= \text{Sample size} = 12 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{12.9 + 12.5 + 11 + ... + 11.3}{10} = 12.17 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{14.1 + 14.1 + 14.2 + .... + 19.0}{12} = 14.875 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n-1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(12.9 - 12.17)^2 + ... + (11.3 - 12.17)^2}{10-1}} \\ &= 1.0563 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(14.1 - 14.875)^2 + .... + (19.0 - 14.875)^2}{12-1}} \\ &= 2.2079 \end{aligned}$$

Step 2

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{1.0563^2}{10} + \frac{2.2079^2}{12}\right)^2}{\frac{(1.0563^2/10)^2}{10 - 1} + \frac{(2.2079^2/12)^2}{12 - 1}} \approx 16$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 16$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.120$$

The rejection region then contains all values smaller than $-2.120$ and all values greater than $2.120$,thus we reject $H_0$ when $t < -2.120$ or $t > 2.120$.

Step 3

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{12.17 - 14.875 - 0}{\sqrt{\frac{1.0563^2}{10} + \frac{2.2079^2}{12}}} \approx -3.759$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-3.759 < -2.120 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean times.

Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 403

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 7 \\ n_2 &= \text{Sample size} = 8 \\ \alpha &= \text{Significance level} = 0.01 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{138 + 121 + 88 + ... + 96}{7} = 117.7143 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{128 + 134 + 152 + ... + 120}{8} = 125.1250 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(138 - 117.7143)^2 + ... + (96 - 117.7143)^2}{7 - 1}} \\ &= 19.9141 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(128 - 125.1250)^2 + ... + (120 - 125.1250)^2}{8 - 1}} \\ &= 15.0943 \end{aligned}$$

Step 2

Given claim: Mean is higher for reduced price

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 \geq \mu_2$$

$$H_1 : \mu_1 < \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Left-tailed

The rejection region of a left-tailed test with $\alpha = 0.01$ contains all t-value below the t-value $-t_0$ that has a probability of 0.01 to its left.

$$P(t < -t_0) = 0.01$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 7 + 8 - 2 = 13$ and in the column with $\alpha = 0.01$ (which is the negation of the critical value in the table).

$$t = 2.650$$

The rejection region then contains all values smaller than $-2.650$ and thus we reject $H_0$ when the t-value is smaller than $-2.650$.

Step 3

Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(7 - 1)19.9141^2 + (8 - 1)15.0943^2}{7 + 8 - 2}} \approx 17.4847$$

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{117.7143 - 125.1250}{17.4847\sqrt{\frac{1}{7} + \frac{1}{8}}} \approx -0.819$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -0.819 > -2.650 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the price reduction resulted in an increase in sales.

Result

No

# Exercise 50



**PROBLEM STATEMENT**

Chapter 11, Page 403

Step 1

Given:

$$\alpha = \text{Significance level} = 0.01$$
$$n = \text{Sample size} = 8$$

**Step 2**

Given claim: Mean is positive (reduced)

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.
The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis are the opposite of each other. The null hypothesis is right-tailed to contain the value mentioned in the claim.
If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.
If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

$$H_0 : \mu_d \leq 0$$
Right-tailed
$$H_a : \mu_d > 0$$

The rejection region of a right-tailed test with $\alpha = 0.01$ contains all t-value above the t-value $t_0$ that has a probability of 0.01 to its right.

$$P(t > t_0) = 0.01$$

Let us determine the difference between each pair of data values:

| Population 1 | Population 2 | Difference d |
|---|---|---|
| 5 | 3 | $5 - 3 = 2$ |
| 7 | 7 | $7 - 7 = 0$ |
| 6 | 7 | $6 - 7 = -1$ |
| 4 | 0 | $4 - 0 = 4$ |
| 8 | 4 | $8 - 4 = 4$ |
| 9 | 6 | $9 - 6 = 3$ |
| 8 | 8 | $8 - 8 = 0$ |
| 10 | 2 | $10 - 2 = 8$ |

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 8 - 1 = 7$ and in the column with $\alpha = 0.01$.

$$t = 2.998$$

The rejection region then contains all values greater than $2.998$ and thus we reject $H_0$ when the t-value is larger than 2.998.

Determine the value of the test statistic:

$$t = \frac{\overline{d}}{s_d/\sqrt{n}} = \frac{2.5}{2.9277/\sqrt{8}} \approx 2.415$$

We reject the null hypothesis $H_0$ when the value of the test statistic is in the rejection region.

$$t = 2.415 < 2.998 \Rightarrow \text{Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the modification reduced the number of traffic accidents.

The mean is the sum of all values divided by the number of values:

**Result**

$$\overline{d} = \frac{2 + 0 - 1 + 4 + 4 + 3 + 0 + 8}{8} = \frac{20}{8} = 2.5$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_d = \sqrt{\frac{(2 - 2.5)^2 + (0 - 2.5)^2 + (-1 - 2.5)^2 + (4 - 2.5)^2 + (4 - 2.5)^2 + (3 - 2.5)^2 + (0 - 2.5)^2 + (8 - 2.5)^2}{8 - 1}} \approx 2.9277$$

**PROBLEM STATEMENT**

Chapter 11, Page 404

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 8$$

Given claim: Mean is positive (declined)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu_d \leq 0$$
$$H_a : \mu_d > 0$$

Let us determine the difference between each pair of data values:

| **Population 1** | **Population 2** | **Difference** d |
|---|---|---|
| 6 | 5 | $6-5=1$ |
| 6 | 2 | $6-2=4$ |
| 7 | 1 | $7-1=6$ |
| 7 | 3 | $7-3=4$ |
| 4 | 3 | $4-3=1$ |
| 3 | 6 | $3-6=-3$ |
| 5 | 3 | $5-3=2$ |
| 6 | 7 | $6-7=-1$ |

The mean is the sum of all values divided by the number of values:

$$\overline{d} = \frac{\begin{matrix}1+4+6+4\\+1-3+2-1\end{matrix}}{8} = \frac{14}{8} = 1.75$$

The variance is the sum of squared deviations from the mean divided by $n-1$. The standard deviation is the square root of the variance:

$$s_d = \sqrt{\frac{\begin{matrix}(1-1.75)^2+(4-1.75)^2+(6-1.75)^2+(4-1.75)^2\\+(1-1.75)^2+(-3-1.75)^2+(2-1.75)^2+(-1-1.75)^2\end{matrix}}{8-1}} \approx 2.9155$$

Step 2

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value $t_0$ that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 8 - 1 = 7$ and in the column with $\alpha = 0.05$.

$$t = 1.895$$

The rejection region then contains all values greater than $1.895$ and thus we reject $H_0$ when the t-value is larger than 1.895.

Determine the value of the test statistic:

$$t = \frac{\overline{d}}{s_d/\sqrt{n}} = \frac{1.75}{2.9155/\sqrt{8}} \approx 1.698$$

We reject the null hypothesis $H_0$ when the value of the test statistic is in the rejection region.

$$t = 1.698 < 1.895 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the number of absences has declined.

Step 3

### P-value

The P-value is the probability of obtaining the value of the test statistic, or a value more extreme, assuming that the null hypothesis is true. The P-value is the number (or interval) in the column title of the Student's T table in the appendix that contains the t-value $1.698$ in the row $df = n - 1 = 8 - 1 = 7$ (one-tailed):

$$0.05 < P < 0.10$$

Note: When using technology, then you obtain the P-value $0.06666$.

🎯 **Result**

No, $0.05 < P < 0.10$

**PROBLEM STATEMENT**

Chapter 11, Page 404

Step 1

Given:

$$\alpha = \text{Significance level} = 0.10$$
$$n = \text{Sample size} = 15$$

Given claim: Mean is nonzero (difference)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu_d = 0$$
$$H_a : \mu_d \neq 0$$

Step 2

Let us determine the difference between each pair of data values:

| Population 1 | Population 2 | Difference d |
|---|---|---|
| 2090 | 1610 | $2090 - 1610 = 480$ |
| 1683 | 1247 | $1683 - 1247 = 436$ |
| 1402 | 2327 | $1402 - 2327 = -925$ |
| 1830 | 1367 | $1830 - 1367 = 463$ |
| 990 | 1461 | $990 - 1461 = -531$ |
| 697 | 1789 | $697 - 1789 = -1092$ |
| 1741 | 1621 | $1741 - 1621 = 120$ |
| 1129 | 1914 | $1129 - 1914 = -785$ |
| 1018 | 1956 | $1018 - 1956 = -938$ |
| 1881 | 1772 | $1881 - 1772 = 109$ |
| 1571 | 1375 | $1571 - 1375 = 196$ |
| 874 | 1527 | $874 - 1527 = -653$ |
| 1579 | 1767 | $1579 - 1767 = -188$ |
| 1577 | 1636 | $1577 - 1636 = -59$ |
| 860 | 1188 | $860 - 1188 = -328$ |

The mean is the sum of all values divided by the number of values:

**Result**

$$\overline{d} = \frac{480 + 436 - 925 + ... - 328}{15} = \frac{-3695}{15} = -246.3333$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_d = \sqrt{\frac{(480 - (-246.3333))^2 + ... + (-328 - (-246.3333))^2}{15 - 1}} \approx 546.9590$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.10$ contains all t-values below the t-value $-t_0$ that has a probability of $0.10/2 = 0.05$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.10/2 = 0.05$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.10}{2} = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 15 - 1 = 14$ and in the column with $\alpha = 0.10$ (two-tailed).

$$t = 1.761$$

The rejection region then contains all values smaller than $-1.761$ and all values greater than $1.761$, thus we reject $H_0$ when $t < -1.761$ or $t > 1.761$.

Determine the value of the test statistic:

$$t = \frac{\overline{d}}{s_d/\sqrt{n}} = \frac{-246.3333}{546.9590/\sqrt{15}} \approx -1.744$$

We reject the null hypothesis $H_0$ when the value of the test statistic is in the rejection region.

$$-1.761 < -1.744 < 1.761 \Rightarrow \text{Fail to reject } H_0$$

No

There is not sufficient evidence to support the claim that there is a difference in the amounts quoted.

**PROBLEM STATEMENT**

Chapter 11, Page 404

Step 1

Given:

$$n_1 = \text{Sample size} = 15$$
$$n_2 = \text{Sample size} = 25$$
$$\overline{x}_1 = \text{Sample mean} = 150000$$
$$\overline{x}_2 = \text{Sample mean} = 180000$$
$$s_1 = \text{Sample standard deviation} = 40000$$
$$s_2 = \text{Sample standard deviation} = 30000$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

Step 2

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 15 + 25 - 2 = 38$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.024$$

The rejection region then contains all values smaller than $-2.024$ and all values greater than $2.024$,thus we reject $H_0$ when $t < -2.024$ or $t > 2.024$.

Step 3

Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(15 - 1)40000^2 + (25 - 1)30000^2}{15 + 25 - 2}} \approx 34028$$

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{150000 - 180000}{34028\sqrt{\frac{1}{15} + \frac{1}{25}}} \approx -2.699$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -2.699 < -2.024 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the population means are different.

🎯 Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 404

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 12 \\ n_2 &= \text{Sample size} = 12 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{3+1+0+\ldots+4}{12} = 2.3333 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{1+1+0+\ldots+0}{12} = 0.5833 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n-1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum(x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(3-2.3333)^2+\ldots+(4-2.3333)^2}{12-1}} \\ &= 1.3707 \\ s_2 &= \sqrt{\frac{\sum(x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(1-0.5833)^2+\ldots+(0-0.5833)^2}{12-1}} \\ &= 0.5149 \end{aligned}$$

Step 2

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 12 + 12 - 2 = 22$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.074$$

The rejection region then contains all values smaller than $-2.074$ and all values greater than $2.074$,thus we reject $H_0$ when $t < -2.074$ or $t > 2.074$.

Step 3

Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} = \sqrt{\frac{(12 - 1)1.3707^2 + (12 - 1)0.5149^2}{12 + 12 - 2}} \approx 1.0354$$

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{2.3333 - 0.5833}{1.0354\sqrt{\frac{1}{12} + \frac{1}{12}}} \approx 4.140$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = 4.140 > 2.074 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the data show a difference in the taste ratings.

Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 405

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05$$
$$n = \text{Sample size} = 8$$

Given claim: Mean is positive (lower after)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu_d \leq 0$$
$$H_a : \mu_d > 0$$

Let us determine the difference between each pair of data values:

| **Population 1** | **Population 2** | **Difference** d |
|---|---|---|
| 6.6 | 6.8 | $6.6 - 6.8 = -0.2$ |
| 6.5 | 2.4 | $6.5 - 2.4 = 4.1$ |
| 9.0 | 7.4 | $9.0 - 7.4 = 1.6$ |
| 10.3 | 8.5 | $10.3 - 8.5 = 1.8$ |
| 11.2 | 8.1 | $11.2 - 8.1 = 3.1$ |
| 8.1 | 6.1 | $8.1 - 6.1 = 2.0$ |
| 6.3 | 3.4 | $6.3 - 3.4 = 2.9$ |
| 11.6 | 2.0 | $11.6 - 2.0 = 9.6$ |

The mean is the sum of all values divided by the number of values:

$$\overline{d} = \frac{\begin{array}{c}-0.2 + 4.1 + 1.6 + 1.8 \\ +3.1 + 2.0 + 2.9 + 9.6\end{array}}{8} = \frac{24.9}{8} = 3.1125$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_d = \sqrt{\frac{\begin{array}{c}(-0.2 - 3.1125)^2 + (4.1 - 3.1125)^2 + (1.6 - 3.1125)^2 \\ +(1.8 - 3.1125)^2 + (3.1 - 3.1125)^2 + (2.0 - 3.1125)^2 \\ +(2.9 - 3.1125)^2 + (9.6 - 3.1125)^2\end{array}}{8 - 1}} \approx 2.9113$$

Step 2

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value $t_0$ that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 8 - 1 = 7$ and in the column with $\alpha = 0.05$.

$$t = 1.895$$

The rejection region then contains all values greater than $1.895$ and thus we reject $H_0$ when the t-value is larger than 1.895.

Determine the value of the test statistic:

$$t = \frac{\overline{d}}{s_d/\sqrt{n}} = \frac{3.1125}{2.9113/\sqrt{8}} \approx 5.040$$

We reject the null hypothesis $H_0$ when the value of the test statistic is in the rejection region.

$$t = 5.040 > 1.895 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the contamination measurements are lower after use of the new soap.

Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 405

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 5 \\ n_2 &= \text{Sample size} = 5 \\ \alpha &= \text{Significance level} = 0.10 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{17.16 + 17.08 + 15.51 + 8.43 + 25.15}{5} = 16.666 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{15.8 + 16.28 + 16.21 + 17.97 + 7.77}{5} = 14.806 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n-1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(17.16 - 16.666)^2 + \ldots + (25.15 - 16.666)^2}{5-1}} \\ &= 5.9490 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(15.8 - 14.806)^2 + \ldots + (7.77 - 14.806)^2}{5-1}} \\ &= 4.0202 \end{aligned}$$

Step 2

Given claim: Mean is higher on NYSE (big board)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 \leq \mu_2$$

$$H_1 : \mu_1 > \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Right-tailed

The rejection region of a right-tailed test with $\alpha = 0.10$ contains all t-value above the t-value $t_0$ that has a probability of 0.10 to its right.

$$P(t > t_0) = 0.10$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 5 + 5 - 2 = 8$ and in the column with $\alpha = 0.10$.

$$t = 1.397$$

The rejection region then contains all values greater than $1.397$ and thus we reject $H_0$ when the t-value is larger than 1.397.

Step 3

Determine the pooled standard deviation:

$$s_p = \sqrt{\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}} = \sqrt{\frac{(5-1)5.9490^2 + (5-1)4.0202^2}{5+5-2}} \approx 5.0770$$

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{16.666 - 14.806}{5.0770\sqrt{\frac{1}{5}+\frac{1}{5}}} \approx 0.579$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = 0.579 < 1.397 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim that the annual rates of return are higher on the big board.

Result

No

Chapter 11

## Exercise 57

Step 1

4. Step & Result Cards Not found

**PROBLEM STATEMENT**

Chapter 11, Page 405

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 25 \\ n_2 &= \text{Sample size} = 28 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{89 + 115 + 93 + ... + 64}{25} = 86.24 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{128 + 110 + 81 + ... + 124}{28} = 92.0357 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(89 - 86.24)^2 + .... + (64 - 86.24)^2}{25 - 1}} \\ &= 23.4330 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(128 - 92.0357)^2 + .... + (124 - 92.0357)^2}{28 - 1}} \\ &= 24.1193 \end{aligned}$$

Step 2

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n_1 + n_2 - 2 = 25 + 28 - 2 = 51$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.008$$

The rejection region then contains all values smaller than $-2.008$ and all values greater than $2.008$,thus we reject $H_0$ when $t < -2.008$ or $t > 2.008$.

Step 3

Determine the pooled standard deviation:

$$\begin{aligned} s_p &= \sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}} \\ &= \sqrt{\frac{(25-1)23.4330^2+(28-1)24.1193^2}{25+28-2}} \\ &\approx 23.7988 \end{aligned}$$

Determine the test statistic:

$$t = \frac{\overline{x}_1-\overline{x}_2}{s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} = \frac{86.24-92.0357}{23.7988\sqrt{\frac{1}{25}+\frac{1}{28}}} \approx -0.885$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -0.885 > -2.008 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean number of cars in the two lots.

Result

No

**PROBLEM STATEMENT**

Chapter 11, Page 406

Step 1

Given:

$$\alpha = \text{Significance level} = 0.05 \quad \text{(assumption)}$$
$$n = \text{Sample size} = 20$$

Given claim: Mean is positive (less now than five years ago)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to contain the value mentioned in the claim.

$$H_0 : \mu_d \leq 0$$
$$H_a : \mu_d > 0$$

Step 2

Let us determine the difference between each pair of data values:

| Population 1 | Population 2 | Difference d |
|---|---|---|
| 17 | 10 | $17 - 10 = 7$ |
| 20 | 39 | $20 - 39 = -19$ |
| 29 | 37 | $29 - 37 = -8$ |
| 43 | 27 | $43 - 27 = 16$ |
| 36 | 12 | $36 - 12 = 24$ |
| 43 | 41 | $43 - 41 = 2$ |
| 45 | 24 | $45 - 24 = 21$ |
| 19 | 26 | $19 - 26 = -7$ |
| 49 | 28 | $49 - 28 = 21$ |
| 49 | 26 | $49 - 26 = 23$ |
| 35 | 32 | $35 - 32 = 3$ |
| 16 | 32 | $16 - 32 = -16$ |
| 23 | 21 | $23 - 21 = 2$ |
| 33 | 12 | $33 - 12 = 21$ |
| 44 | 40 | $44 - 40 = 4$ |
| 44 | 42 | $44 - 42 = 2$ |
| 28 | 22 | $28 - 22 = 6$ |
| 29 | 19 | $29 - 19 = 10$ |
| 39 | 35 | $39 - 35 = 4$ |
| 22 | 12 | $22 - 12 = 10$ |

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

$$\text{Right-tailed}$$

The rejection region of a right-tailed test with $\alpha = 0.05$ contains all t-value above the t-value $t_0$ that has a probability of 0.05 to its right.

$$P(t > t_0) = 0.05$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = n - 1 = 20 - 1 = 19$ and in the column with $\alpha = 0.05$.

$$t = 1.729$$

The rejection region then contains all values greater than $1.729$ and thus we reject $H_0$ when the t-value is larger than 1.729.

Determine the value of the test statistic:

$$t = \frac{\overline{d}}{s_d/\sqrt{n}} = \frac{6.3}{12.4778/\sqrt{20}} \approx 2.258$$

We reject the null hypothesis $H_0$ when the value of the test statistic is in the rejection region.

$$t = 2.258 > 1.729 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that the percent is less now than five years ago.

The mean is the sum of all values divided by the number of values:

$$\overline{d} = \frac{7 - 19 - 8 + ... + 10}{20} = \frac{126}{20} = 6.3$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s_d = \sqrt{\frac{(7 - 6.3)^2 + .... + (10 - 6.3)^2}{20 - 1}} \approx 12.4778$$

🎯 **Result**

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 406

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 15 \\ n_2 &= \text{Sample size} = 12 \\ \alpha &= \text{Significance level} = 0.01 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{4+4+4+\ldots+75}{15} = 39.4 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{200+100+200+\ldots.+300}{12} = 187.5 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n-1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum(x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(4-39.4)^2+\ldots+(75-39.4)^2}{15-1}} \\ &= 53.2498 \\ s_2 &= \sqrt{\frac{\sum(x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(200-187.5)^2+\ldots+(300-187.5)^2}{12-1}} \\ &= 48.2654 \end{aligned}$$

Step 2

Given claim: The means is lower for the first four rows (higher odds)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 \geq \mu_2$$

$$H_1 : \mu_1 < \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Left-tailed

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{53.2498^2}{15} + \frac{48.2654^2}{12}\right)^2}{\frac{(53.2498^2/15)^2}{15 - 1} + \frac{(48.2654^2/12)^2}{12 - 1}} \approx 24$$

The rejection region of a left-tailed test with $\alpha = 0.01$ contains all t-value below the t-value $-t_0$ that has a probability of 0.01 to its left.

$$P(t < -t_0) = 0.01$$

Determine the critical value from the Student's T distribution table in the appendix in the row with $df = 24$ and in the column with $\alpha = 0.01$ (which is the negation of the critical value in the table).

$$t = -2.492$$

The rejection region then contains all values smaller than $-2.492$ and thus we reject $H_0$ when the t-value is smaller than $-2.492$.

Step 3

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{39.4 - 187.5 - 0}{\sqrt{\frac{53.2498^2}{15} + \frac{48.2654^2}{12}}} \approx -7.566$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -7.566 < -2.492 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that starting in the first five rows significantly increases the odds of winning, in contrast to the last four rows.

Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 406

Step 1

Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 20 \\ n_2 &= \text{Sample size} = 13 \\ \alpha &= \text{Significance level} = 0.01 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{4+4+4+\ldots+150}{20} = 60.9 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{8+20+200+\ldots+300}{13} = 159.8462 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n-1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum(x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(4-60.9)^2+\ldots+(150-60.9)^2}{20-1}} \\ &= 63.1905 \\ s_2 &= \sqrt{\frac{\sum(x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(8-159.8462)^2+\ldots+(300-159.8462)^2}{13-1}} \\ &= 82.0588 \end{aligned}$$

Step 2

Given claim: The means differ (change)

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{63.1905^2}{20} + \frac{82.0588^2}{13}\right)^2}{\frac{(63.1905^2/20)^2}{20 - 1} + \frac{(82.0588^2/13)^2}{13 - 1}} \approx 21$$

The rejection region of a two-tailed test with $\alpha = 0.01$ contains all t-values below the t-value $-t_0$ that has a probability of $0.01/2 = 0.005$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.01/2 = 0.005$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.01}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 21$ and in the column with $\alpha = 0.01$ (two-tailed).

$$t = 2.831$$

The rejection region then contains all values smaller than $-2.831$ and all values greater than $2.831$,thus we reject $H_0$ when $t < -2.831$ or $t > 2.831$.

Step 3

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{60.9 - 159.8462 - 0}{\sqrt{\frac{63.1905^2}{20} + \frac{82.0588^2}{13}}} \approx -3.694$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -3.694 < -2.831 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim that a car number of 20 or below significantly change the odds of winning.

Result

Yes

**PROBLEM STATEMENT**

Chapter 11, Page 406

Step 1

(a) Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 38 \\ n_2 &= \text{Sample size} = 67 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{224 + 206 + 179 + ... + 175.6}{38} = 202.7974 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{243.7 + 246 + 270.8 + ... + 139.9}{67} = 231.4851 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(224 - 202.7974)^2 + .... + (175.6 - 202.7974)^2}{38 - 1}} \\ &= 33.7051 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(243.7 - 231.4851)^2 + ... + (139.9 - 231.4851)^2}{67 - 1}} \\ &= 50.5693 \end{aligned}$$

Step 2

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{33.7051^2}{38} + \frac{50.5693^2}{67}\right)^2}{\frac{(33.7051^2/38)^2}{38 - 1} + \frac{(50.5693^2/67)^2}{67 - 1}} \approx 100$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 100$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 1.984$$

The rejection region then contains all values smaller than $-1.984$ and all values greater than $1.984$,thus we reject $H_0$ when $t < -1.984$ or $t > 1.984$.

Step 3

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{202.7974 - 231.4851 - 0}{\sqrt{\frac{33.7051^2}{38} + \frac{50.5693^2}{67}}} \approx -3.477$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -3.477 < -1.984 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean selling price of homes with a pool and homes without a pool.

Step 4

(b) Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 34 \\ n_2 &= \text{Sample size} = 71 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

## Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{206 + 207.5 + 147.4 + ... + 139.9}{34} = 185.45 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{224 + 179 + 198.9 + ... + 189.4}{71} = 238.1761 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(206 - 185.45)^2 + .... + (139.9 - 185.45)^2}{34 - 1}} \\ &= 28.0046 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(224 - 238.1761)^2 + ... + (189.4 - 238.1761)^2}{71 - 1}} \\ &= 50.5693 \end{aligned}$$

Step 6

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{185.45 - 238.1761 - 0}{\sqrt{\frac{28.0046^2}{34} + \frac{44.8765^2}{71}}} \approx -7.352$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -7.352 < -1.975 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean selling price of homes with an attached garage and homes without an attached garage.

Step 7

(c) Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 15 \\ n_2 &= \text{Sample size} = 20 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

## Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{147.4 + 192.6 + ... + 189.4}{15} = 196.9133 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{207.5 + 154.3 + ... + 281.3}{20} = 227.45 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n-1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(147.4-196.9133)^2 + .... + (189.4-196.9133)^2}{15-1}} \\ &= 35.7841 \\ s_2 &= \sqrt{\frac{\sum (x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(207.5-227.45)^2 + ... + (281.3-227.45)^2}{20-1}} \\ &= 43.0744 \end{aligned}$$

Step 9

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{196.9133 - 227.45 - 0}{\sqrt{\frac{35.7841^2}{15} + \frac{43.0744^2}{20}}} \approx -2.288$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -2.288 < -2.037 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean selling price of homes in Township 1 and Township 2.

Step 10

(d) Order the data values from smallest to largest:

125, 125.9, 139.9, 147.4, 154.3, 155.4, 164.1, 166.2, 166.5, 171.6, 172.4, 172.7, 173.1, 173.6, 173.6, 175, 175.6, 176, 176.3, 177.1, 179, 179, 180.4, 182.4, 182.7, 186.7, 187, 188.1, 188.3, 188.3, 188.3, 188.3, 189.4, 190.9, 192.2, 192.6, 192.9, 194.4, 198.3, 198.9, 199, 199.8, 205.1, 206, 207.1, 207.5, 207.5, 209, 209.3, 209.3, 209.7, 209.7, 213.6, 216, 216.8, 217.8, 220.9, 221.1, 221.5, 222.1, 224, 224.8, 227.1, 227.1, 228.4, 232.2, 233, 234, 236.4, 236.8, 240, 242.1, 243.7, 244.6, 245.4, 246, 246.1, 247.7, 251.4, 252.3, 253.2, 254.3, 257.2, 263.1, 263.2, 266.6, 269.2, 269.9, 270.8, 271.8, 273.2, 281.3, 289.8, 292.4, 293.7, 294, 294.3, 294.5, 307.8, 310.8, 312.1, 312.1, 326.3, 327.2, 345.3

The **median** is the middle value of the sorted data set. Since there are 105 data values in the data set, the mean is the 53rd data value in the sorted data set.

$$m = 213.6$$

Step 11

Given:

$$n_1 = \text{Sample size} = 38$$
$$n_2 = \text{Sample size} = 67$$
$$x_1 = \text{Number of successes} = 28$$
$$x_2 = \text{Number of successes} = 24$$
$$\alpha = \text{Significance level} = 0.05$$

Given claim: Difference in the proportions

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis includes the equality. The alternative hypothesis states the opposite of the null hypothesis.

$$H_0 : \pi_1 = \pi_2$$
$$H_a : \pi_1 \neq \pi_2$$

Step 12

If the alternative hypothesis $H_a$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_a$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_a$ contains $\neq$, then the test is two-tailed.

Two-tailed test

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all z-values below the z-value $-z_0$ that has a probability of $0.05/2 = 0.025$ to its left and all z-values above the z-value $z_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(z < z_0) = P(z > z_0) = \frac{0.05}{2} = 0.025$$

Let us determine the z-score that corresponds with a probability of $1 - 0.5 - 0.025 = 0.475$ in the normal probability table of the appendix (the z-score $z_0$ is then the negative of this z-score). We note that the probability 0.475 is closest to 0.4750, which is given in the row starting with "1.9" and in the column starting with ".06", thus the corresponding z-score is $1.9 + .06 = 1.96$.

$$z_0 = \pm 1.96$$

Thus the rejection region then contains all z-scores smaller than $-1.96$ and all z-scores larger than 1.96, thus we reject $H_0$ when the z-score is smaller than $-1.96$ or the z-score is larger than 1.96.

Step 13

The sample proportion is the number of successes divided by the sample size:

$$\hat{p}_1 = \frac{x_1}{n_1} = \frac{28}{38} \approx 0.7368$$
$$\hat{p}_2 = \frac{x_2}{n_2} = \frac{24}{67} \approx 0.3582$$
$$\hat{p}_c = \frac{x_1 + x_2}{n_1 + n_2} = \frac{28 + 24}{38 + 67} = \frac{52}{105} \approx 0.4952$$

Determine the value of the test statistic:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c(1 - \hat{p}_c)}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{0.7368 - 0.3582}{\sqrt{0.4952(1 - 0.4952)}\sqrt{\frac{1}{38} + \frac{1}{67}}} \approx 3.73$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$z = 3.73 > 1.96 \Rightarrow \text{ Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the proportion of homes with a pool for those that sold at or above the median price versus those that sold for less than the median price.

Step 14

(e) We need to summarize the results of part (a) to (d).

To all real estate agents who sell property in Goodyear:

Based on the results of a study, we concluded that a pool and a garage both affect the selling price of a house. Moreover, the selling prices of homes in Township 1 and Township 2 differed as well.

Kind regards,

Sarah Schrijvers

🎯 **Result**

(a) Yes

(b) Yes

(c) Yes

(d) Yes

(e) Summarize the results of part (a) to (d)

**PROBLEM STATEMENT**

Chapter 11, Page 407

Step 1

(a) Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 16 \\ n_2 &= \text{Sample size} = 14 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{36.8 + 437. + 48.7 + ... + 149.4}{16} = 84.3438 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{62.3 + 63.3 + 65.3 + ... + 201.5}{14} = 93.2786 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n-1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(36.8-84.3438)^2 + ... + (149.4-84.3438)^2}{16-1}} \\ &= 31.0924 \\ s_2 &= \sqrt{\frac{\sum (x-\overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(62.3-93.2786)^2 + ... + (201.5-93.2786)^2}{14-1}} \\ &= 37.4496 \end{aligned}$$

Step 2

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{31.0924^2}{16} + \frac{37.4496^2}{14}\right)^2}{\frac{(31.0924^2/16)^2}{16 - 1} + \frac{(37.4496^2/14)^2}{14 - 1}} \approx 25$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 25$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.060$$

The rejection region then contains all values smaller than $-2.060$ and all values greater than $2.060$,thus we reject $H_0$ when $t < -2.060$ or $t > 2.060$.

Step 3

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{84.3438 - 93.2786 - 0}{\sqrt{\frac{31.0924^2}{16} + \frac{37.4496^2}{14}}} \approx -0.705$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-2.06 < -0.705 < 2.060 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean salary of teams in the American League versus teams in the National League.

Step 4

(b) Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 16 \\ n_2 &= \text{Sample size} = 14 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

## Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{1.46 + 1.92 + 1.58 + ... + 3.15}{16} = 2.5713 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{1.41 + 1.87 + 2.42 + ... + 3.72}{14} = 2.3064 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(1.46 - 2.5713)^2 + ... + (3.15 - 2.5713)^2}{16 - 1}} \\ &= 0.7388 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(1.41 - 2.3064)^2 + ... + (3.72 - 2.3064)^2}{14 - 1}} \\ &= 0.6463 \end{aligned}$$

Step 6

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{2.5713 - 2.3064 - 0}{\sqrt{\frac{0.7388^2}{16} + \frac{0.6463^2}{14}}} \approx 1.048$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-2.048 < 1.048 < 2.048 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean home attendance of teams in the American League versus teams in the National League.

Step 7

(c) Given:

$$\begin{aligned} n_1 &= \text{Sample size} = 10 \\ n_2 &= \text{Sample size} = 10 \\ \alpha &= \text{Significance level} = 0.05 \end{aligned}$$

## Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\begin{aligned} \overline{x}_1 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{36.8 + 43.7 + 48.7 + .... + 70.5}{10} = 58.62 \\ \overline{x}_2 &= \frac{\sum_{i=1}^{n} x_i}{n} = \frac{98.9 + 100.4 + 103 + ... + 201.5}{10} = 125.16 \end{aligned}$$

The **sample variance** is the sum of squared deviations from the mean divided by $n-1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(36.8 - 58.62)^2 + .... + (70.5 - 58.62)^2}{10-1}} \\ &= 11.4738 \\ s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n-1}} \\ &= \sqrt{\frac{(98.9 - 125.16)^2 + ... + (201.5 - 125.16)^2}{10-1}} \\ &= 31.0800 \end{aligned}$$

Step 9

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{58.62 - 125.16 - 0}{\sqrt{\frac{11.4738^2}{10} + \frac{31.0800^2}{10}}} \approx -6.351$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$t = -6.351 < -2.201 \Rightarrow \text{Reject } H_0$$

There is sufficient evidence to support the claim of a difference in the mean number of wins for the two groups.

Result

(a) No

(b) No

(c) Yes

**PROBLEM STATEMENT**

Chapter 11, Page 407

Step 1

Given:

$$n_1 = \text{Sample size} = 53$$
$$n_2 = \text{Sample size} = 27$$
$$\alpha = \text{Significance level} = 0.05$$

### Descriptive measures

The **mean** is the sum of all values divided by the number of values:

$$\overline{x}_1 = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{329 + 503 + 505 + ... + 426}{53} = 454.7736$$
$$\overline{x}_2 = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{466 + 359 + 427 + ... + 475}{27} = 441.4815$$

The **sample variance** is the sum of squared deviations from the mean divided by $n - 1$. The **sample standard deviation** is the square root of the population sample.

$$\begin{aligned} s_1 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(329 - 454.7736)^2 + ... + (426 - 454.7736)^2}{53 - 1}} \\ &= 52.9363 \end{aligned}$$

$$\begin{aligned} s_2 &= \sqrt{\frac{\sum (x - \overline{x})^2}{n - 1}} \\ &= \sqrt{\frac{(466 - 441.4815)^2 + ... + (475 - 441.4815)^2}{27 - 1}} \\ &= 55.0576 \end{aligned}$$

Step 2

Given claim: The means differ

The claim is either the null hypothesis or the alternative hypothesis. The null hypothesis and the alternative hypothesis state the opposite of each other. The null hypothesis needs to include an equality.

$$H_0 : \mu_1 = \mu_2$$

$$H_1 : \mu_1 \neq \mu_2$$

If the alternative hypothesis $H_1$ contains $<$, then the test is left-tailed.

If the alternative hypothesis $H_1$ contains $>$, then the test is right-tailed.

If the alternative hypothesis $H_1$ contains $\neq$, then the test is two-tailed.

Two-tailed

Determine the degrees of freedom (rounded down to the nearest integer):

$$\Delta = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}} = \frac{\left(\frac{52.9363^2}{53} + \frac{55.0576^2}{27}\right)^2}{\frac{(52.9363^2/53)^2}{53 - 1} + \frac{(55.0576^2/27)^2}{27 - 1}} \approx 50$$

The rejection region of a two-tailed test with $\alpha = 0.05$ contains all t-values below the t-value $-t_0$ that has a probability of $0.05/2 = 0.025$ to its left and all t-values above the t-value $t_0$ that has a probability of $0.05/2 = 0.025$ to its right.

$$P(t < -t_0) = P(t > t_0) = \frac{0.05}{2} = 0.025$$

Determine the critical values from the Student's T distribution table in the appendix in the row with $df = 50$ and in the column with $\alpha = 0.05$ (two-tailed).

$$t = 2.009$$

The rejection region then contains all values smaller than $-2.009$ and all values greater than $2.009$,thus we reject $H_0$ when $t < -2.009$ or $t > 2.009$.

Step 3

Determine the test statistic:

$$t = \frac{\overline{x}_1 - \overline{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{454.7736 - 441.4815 - 0}{\sqrt{\frac{52.9363^2}{53} + \frac{55.0576^2}{27}}} \approx 1.034$$

When the value of the test statistic is in the rejection region, then we reject the null hypothesis $H_0$:

$$-2.009 < 1.034 < 2.009 \Rightarrow \text{ Fail to reject } H_0$$

There is not sufficient evidence to support the claim of a difference in the mean maintenance cost for the diesel versus the gasoline buses.

Result

No